Requirements Specification Document - Final

Inclusive and Accessible Hiring

Cool Name

November 22, 2025

Table of Content

1.0 Introduction....7
1.1 Purpose....7
1.2 Project Scope....7
1.3 Glossary of Terms....8
1.4 References....9
1.5 Overview....9
2.0 Overall Description....10
2.1 Product Perspective....10
2.2 Product Features....14
2.3 User Classes and Characteristics....17
2.4 Operating Environment....18
2.5 Design and Implementation Constraints....18
2.6 Assumptions and dependencies....19
3.0 System Features....19
3.1 Resume Parsing and Skill Inference....19
3.1.1 Description and Priority....19
3.1.2 Functional Requirements....20
3.1.3 Use Cases....20
UC-1-1: Uploading a Resume....20
UC-1-1: Sequence Diagram....21
3.2 Voluntary Disclosure and Accommodation Handling....22
3.2.1 Description and Priority....22
3.2.2 Functional Requirements....22
3.2.3 Use Cases....23
UC- 2-1: Submit Voluntary Disclosure....23
UC- 2-1: Sequence Diagram....24
3.3 User Role-Based Interfaces....25
3.3.1 Description and Priority....25
3.3.2 Functional Requirements....26
3.3.3 Use Cases....26
UC-3-1: View Application Status....26
UC-3-1: Sequence Diagram....28
UC-3-1: User Story....28
UC-3-2: Hiring Manager Views Candidate Materials....30
UC-3-2: Sequence Diagram....31
UC-3-3: HR Accesses Audit Logs....32
UC-3-3: Sequence Diagram....33

3.4 Fairness, Transparency, and Explainability....34
3.4.1 Description and Priority....34
3.4.2 Functional Requirements....34
3.4.3 Use Cases....35
UC-4-1: View AI Decision Explanation....35
UC-4-1: Sequence Diagram....36
UC-4-1: User Story....37
UC-4-2: Submit Correction or Feedback on AI Decision....37
UC-4-3: Override AI Decision....38
3.5 Accessibility and Compliance....41
3.5.1 Description and Priority....41
3.5.2 Functional Requirements....41
3.5.3 Use Cases....42
UC-5-1: Run Accessibility Audit....42
UC-5-2: Delete User Data (Right to be Forgotten)....45
UC-5-2: User Story....47
3.6 Context Free Diagram....47
3.7 System-Wide DFD-0....48
4.0 External Interface Requirements....49
4.1 User Interfaces....49
4.1.1 Description and Priority....49
4.1.2 Functional Requirements....49
4.1.3 Use-Case Diagram....50
4.2 Hardware Interfaces....52
4.3 Software Interfaces....53
4.4 Communications Interfaces....54
4.5 Interface Diagrams....56
5.0 Other Non-Functional Requirements....57
5.1 Performance Requirements....57
5.1.1 Description and Priority....57
5.1.2 Non-Functional Requirements....58
5.2 Safety Requirements....58
5.2.1 Description and Priority....58
5.2.2 Non-Functional Requirements....58
5.3 Security Requirements....59
5.3.1 Description and Priority....59
5.3.2 Non-Functional Requirements....59
5.4 Software Quality Attributes....60
5.4.1 Description and Priority....60

5.4.2 Non-Functional Requirements.....60
6.0 Other Requirements.....61
6.1 Database Requirements.....61
6.2 Internationalization Requirements.....61
6.3 Legal Requirements.....62
6.4 Reuse Objectives.....62
7.0 Entity Relationship Diagram.....62
7.1 Entity Relationship Diagram.....63
7.2 Data Dictionary.....63
8.0 Appendix: Issues List.....67
8.1 TBDs.....67
8.2 Information Needed.....68
8.3 Conflicts Awaiting Resolution.....68

Revision History

Name	Date	Reason for Changes	Version
All	07.10.25	Requirements Document 1.0	0.0
Nikhil Dhillon	10.10.25	Updated Section 3 - Req 5.2 as per the feedback by our client team	0.1.0
Jaiman Singh	14.10.25	Updated Section 2 as per the comments and feedback by our client team	0.1.1
Owen Brown	14.10.25	Updated Section 1 as per the comments and feedback by our client team	0.1.2
Amrinder Singh	14.10.25	Updated Section 4 as per the comments and feedback by our client team	0.1.3
Audri	14.10.25	Updated Section 6 as per the comments and feedback by our client team	0.1.4
Cole Knudsen	17.10.25	Updated Section 5 as per the comments and feedback by our client team	0.1.5
Nikhil Dhillon	17.10.25	Added UC 4-1	0.2.0
Audri	17.10.25	Added UC 2-1	0.2.1
Cole Knudsen	17.10.25	Added UC 1-1	0.2.2
Nikhil Dhillon	19.10.25	Added UC 4-2	0.2.3

Owen Brown	**21.10.25**	**Added UC 3-1, 3-2, & 3-3**	**0.2.4**
Nikhil Dhillon	**21.10.25**	**Added Uc 4-3**	**0.2.5**
Jaiman Singh	**21.10.25**	**Made the Use-Case Diagram In Section 4.1**	**0.3**
All	**22.10.25**	**Final formatting changes & review**	**0.4**
All	**22.10.25**	**Final copy (Proofread)**	**RSD 1.0**
Owen Brown	**27.10.25**	**Updated section 1.0 & 8.0 with client team feedback**	**1.1**
Nikhil Dhillon	**27.10.25**	**Updated section 3.0 with client team feedback**	**1.2**
Jaiman Singh	**27.10.25**	**Updated section 2.0 with client team feedback**	**1.3**
Amrinder Singh	**27.10.25**	**Updated section 4.0 with client team feedback**	**1.4**
Audri	**27.10.25**	**Updated section 6.0 with client team feedback**	**1.5**
Cole Knudsen	**27.10.25**	**Updates section 5.0 with client team feedback**	**1.6**
All	**27.10.25**	**Final copy (proof read)**	**RSD 1.1**
Nikhil Dhillon	**04.11.25**	**Added Rationale for section 3**	**1.2.0**
Owen Brown	**04.11.25**	**Helped add Rational**	**1.2.1**

		for section 3	
Jaiman Singh	**04.11.25**	**Added rationale for section 2**	**1.2.2**
Amrinder Singh	**04.11.25**	**Added rationale for section 4**	**1.2.3**
Audri	**04.11.25**	**Added rationale for section 6**	**1.2.4**
Cole Knudsen	**04.11.25**	**Added rationale for section 5**	**1.2.5**
Nikhil Dhillon	**10.11.25**	**Added sequence diagrams for UC 4-1, 4-2, 4-3**	**1.2.0**
Owen Brown	**10.11.25**	**Added sequence diagrams for UC 3-1, 3-2, 3-3**	**1.3.1**
Jaiman Singh	**10.11.25**	**Helped with the ER diagram**	**1.3.3**
Amrinder Singh	**10.11.25**	**Helped with the foreign keys dictionary**	**1.3.4**
Audri	**10.11.25**	**Added the data dictionary**	**1.3.5**
Cole Knudsen	**10.11.25**	**Added sequence diag for UC 1-1**	**1.3.6**
Cole Knudsen	**10.11.25**	**Added the ER diagram**	**1.3.7**
All	**10.11.25**	**Final copy (proof read)**	**RSD 2.0**
Owen Brown	**27.10.25**	**Updated section 1.0 & 8.0 with client team feedback**	**2.0.1**
Nikhil Dhillon	**27.10.25**	**Updated section 3.0 with client team**	**2.0.2**

		feedback	
Jaiman Singh	**27.10.25**	**Updated section 2.0 with client team feedback**	**2.0.3**
Amrinder Singh	**27.10.25**	**Updated section 4.0 with client team feedback**	**2.0.4**
Audri	**27.10.25**	**Updated section 6.0 with client team feedback**	**2.0.5**
Cole Knudsen	**27.10.25**	**Updates section 5.0 with client team feedback**	**2.0.6**
All	**10.11.25**	**Final copy (proof read)**	**RSD 2.1**
Nikhil Dhillon	**16.11.25**	**Added user story and mockups for section 3.4**	**2.1.0**
Owen Brown	**16.11.25**	**Added user story and mockups for section 3.3**	**2.1.1**
Amrinder Singh	**16.11.25**	**Added user story and mockups for section 3.5**	**2.1.2**
Audri	**16.11.25**	**Added user story and mockups for section 3.2**	**2.1.3**
Cole Knudsen	**16.11.25**	**Added user story and mockups for section 3.1**	**2.1.4**
Nikhil Dhillon	**16.11.25**	**Added DFD Level-0**	**2.2.0**
Owen Brown	**16.11.25**	**Added CFD**	**2.3.0**
All	**16.11.25**	**Proofread our sections and the**	**RSD 3.0**

		whole doc	
All	**22.11.25**	**Made changes to respective sections with client team feedback**	**RSD 3.1**

1.0 Introduction

1.1 Purpose

This document specifies the requirements for version 1.0 of the Inclusive and Accessible Hiring Technology. The system aims to provide a fair, transparent, and accessible resume screening and candidate evaluation platform for organizations seeking to reduce bias in recruitment processes.

The requirements document covers the full scope of the system, including functional and non-functional requirements, external interfaces, and compliance considerations. It serves as the baseline for design, implementation, and validation of the product, and will guide communication between stakeholders, designers, and developers.

1.2 Project Scope

The Inclusive and Accessible Hiring Technology is a web-based platform designed to improve fairness, accessibility, and transparency in recruitment. Its primary scope is to assist organizations in evaluating candidates without bias, particularly addressing systemic disadvantages faced by individuals with disabilities and non-linear career paths. The system will leverage Natural Language Processing (NLP) to parse diverse resume formats, Explainable AI (XAI) to ensure transparency in candidate evaluations, and WCAG-compliant interfaces to support accessibility for all users.

The solution will provide tailored functionality for applicants, recruiters, hiring managers, and HR/DEI teams, covering the entire resume screening process from submission to compliance reporting. It will operate as a standalone application.

The key objectives of the system are to:

- Provide bias-aware resume parsing that recognizes non-traditional career histories.
- Enable secure, voluntary disclosure of accommodation needs while protecting candidate privacy.
- Deliver transparent decision-making through audit logs, one-sentence AI explanations, and override mechanisms.
- Ensure accessibility and compliance with international standards such as WCAG 2.1 AA, GDPR, PIPEDA, ADA, and AODA.
- Offer HR/DEI analytics dashboards for diversity tracking and compliance reporting.

By addressing these objectives, the system aligns with corporate diversity, equity, and inclusion strategies, regulatory compliance requirements, and the broader organizational goal of building more inclusive workplaces.

1.3 Glossary of Terms

- **AI (Artificial Intelligence):** The simulation of human intelligence by machines to perform tasks such as decision-making, learning, and problem solving.
- **ATS (Applicant Tracking System):** A recruitment software platform used by organizations to manage job applications and streamline the hiring process. Examples include Greenhouse and Workday.
- **DEI (Diversity, Equity, and Inclusion):** Organizational initiatives and practices that promote fair treatment, access, opportunity, and advancement for all individuals.
- **GDPR (General Data Protection Regulation):** A European Union regulation on data protection and privacy for individuals, requiring secure handling of personal data.
- **PIPEDA (Personal Information Protection and Electronic Documents Act):** A Canadian law governing how private sector organizations collect, use, and disclose personal information.
- **ADA (Americans with Disabilities Act):** U.S. legislation that prohibits discrimination against individuals with disabilities and mandates accessibility in employment, public accommodations, and more.
- **AODA (Accessibility for Ontarians with Disabilities Act):** Ontario legislation requiring organizations to meet accessibility standards for people with disabilities.
- **NLP (Natural Language Processing):** A field of AI that enables computers to understand, interpret, and generate human language, used here to parse resumes.
- **XAI (Explainable AI):** AI methods and techniques that provide human-understandable justifications for decisions or recommendations made by AI systems.
- **WCAG (Web Content Accessibility Guidelines):** International accessibility standards published by W3C to ensure digital content is accessible to people with disabilities.
- **Resume Parsing:** The process of automatically analyzing a resume to extract structured information such as skills, experience, and education.
- **Accommodation Disclosure:** A process by which candidates voluntarily share any required workplace accommodations to support accessibility needs.

- **Audit Log:** A secure, tamper-proof record of system activities, such as AI decisions, user actions, and data access, maintained for transparency and compliance.

- **Role-Based Interfaces:** User interfaces customized for different types of users (e.g., applicants, recruiters, hiring managers, HR/DEI staff), ensuring relevance and accessibility.

1.4 References

- Challenge Statement: *Inclusive & Accessible Hiring Technologies.* Provided as part of course project materials, Fall 2025.

- Request for Proposal (RFP): *Inclusive Hiring System – Requirements and Guidelines.* Provided by client group, Fall 2025.

- Interview Summary with Client Stakeholders, September 2025.

- WCAG 2.1 Web Content Accessibility Guidelines. World Wide Web Consortium (W3C). Available at: https://www.w3.org/WAI/standards-guidelines/wcag/.

- General Data Protection Regulation (GDPR). European Union, Regulation (EU) 2016/679.

- Personal Information Protection and Electronic Documents Act (PIPEDA). Government of Canada, 2000.

- Accessibility for Ontarians with Disabilities Act (AODA). Government of Ontario, 2005.

- Americans with Disabilities Act (ADA). U.S. Department of Justice, 1990.

- IBM. *Explainable AI (XAI) Overview.* IBM Think, 2023. Available at: https://www.ibm.com/think/topics/explainable-ai.

1.5 Overview

This Requirements Document defines the complete set of functional and non-functional requirements for the Inclusive and Accessible Hiring Technology.

- **Section 1.0 Introduction** provides the purpose, scope, glossary, references, and overview of the document.

- **Section 2.0 Overall Description** outlines the product perspective, major features, user classes, operating environment, design constraints, and assumptions.

- **Section 3.0 System Features** details each system feature, its description, priority, and associated functional requirements.

- **Section 4.0 External Interface Requirements** specifies user interfaces, hardware interfaces, software interfaces, and communication requirements.

- **Section 5.0 Other Non-Functional Requirements** describes performance, safety, security, and quality requirements.

- **Section 6.0 Other Requirements** covers database, internationalization, legal, and reuse considerations.

- **Section 7.0 Entity Relationship Diagram** provides our entity relationship diagram for our application as well as its data dictionary.

- **Section 8.0 Appendix** identifies outstanding issues, pending decisions, and conflicts awaiting resolution.

This structure ensures that all stakeholders including designers, developers, clients, and end users—can reference a clear, consistent, and verifiable set of requirements throughout the system's design and development lifecycle.

2.0 Overall Description

2.1 Product Perspective

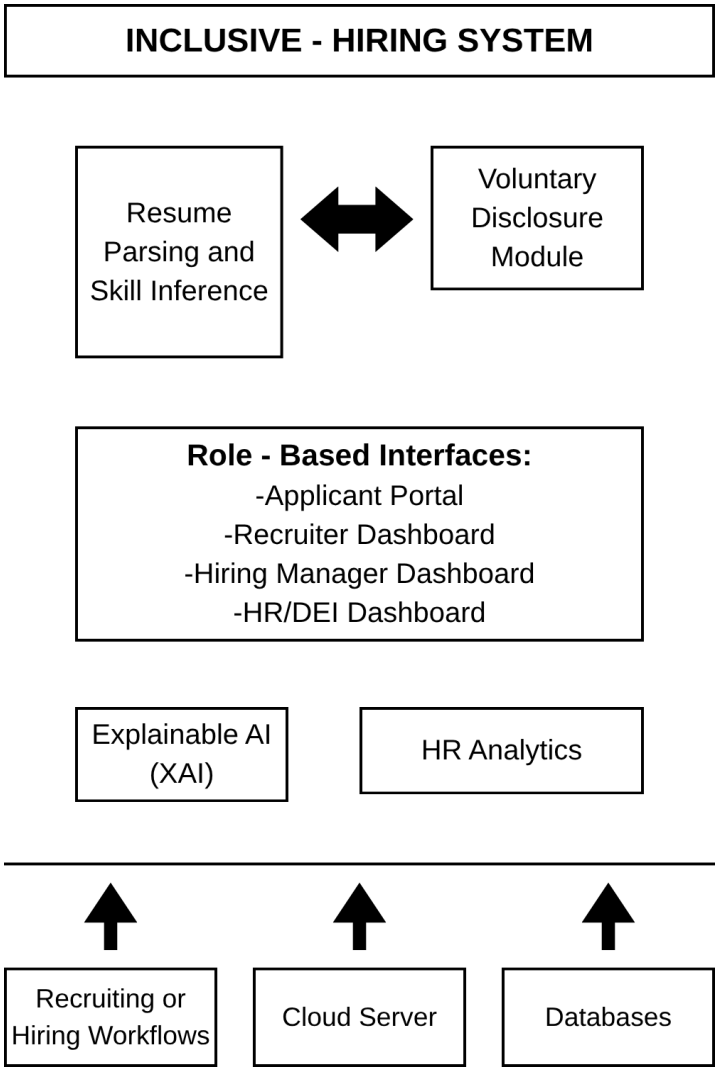
The proposed Inclusive and Accessible Hiring Technology is a new, standalone system designed to integrate with existing recruitment workflows while addressing key shortcomings in current resume screening processes. Unlike legacy Applicant Tracking Systems (ATS) that rely heavily on rigid parsing rules and scoring models, this system is designed with fairness, accessibility, and transparency at its core.

It will leverage **AI-driven Natural Language Processing (NLP)** to parse resumes of varying formats, including those created using assistive technologies, and **Explainable AI (XAI)** techniques to provide transparent decision-making for recruiters and candidates. The system will also include a secure voluntary disclosure mechanism for candidates to share accommodation needs without biasing evaluation.

The solution is designed exclusively as a standalone web application, ensuring focused development and a streamlined user experience. By operating independently from existing ATS platforms like Greenhouse or Workday, it provides organizations with a dedicated tool to enhance inclusivity and efficiency in their hiring processes without requiring platform integration.

A high-level architecture diagram illustrating system components, user interfaces, and integration points is shown in Figure 2-1.

Figure 2-1: Matching Diagram of the Inclusive Hiring System



Explanation for Matching Diagram:

The matching diagram visually represents the relationship between the proposed system features and the project objectives outlined in the RFP and challenge statement. Each feature is mapped to one or more objectives to ensure that the design directly addresses the defined goals of inclusivity, accessibility, fairness, transparency, and compliance.

For example:

- **“Resume Parsing and Skill Inference”** is directly matched with objectives related to skill-based evaluation, career gap recognition, and processing diverse resume formats.

- **“Voluntary Disclosure and Accommodation Handling”** aligns with inclusivity, privacy, and accessibility objectives by providing a secure mechanism for candidates to disclose accommodation needs without bias.

- **“User Role-Based Interfaces”** addresses accessibility standards and user-specific needs for applicants, recruiters, hiring managers, and HR.

- **“Fairness, Transparency, and Explainability”** corresponds to auditability and compliance objectives, ensuring that AI recommendations are explainable and auditable.

- **“Accessibility and Compliance”** matches compliance objectives to meet WCAG standards and data privacy regulations.

This diagram serves as a high-level reference to ensure that each system feature is purposefully designed to meet one or more critical project objectives.

System Context Overview:

The Inclusive and Accessible Hiring System operates as a standalone web application that interacts with several external actors and external systems. The primary external actors include applicants, recruiters, hiring managers, and HR/DEI staff, each interacting with the system through role-specific interfaces. External systems include cloud hosting services, authentication providers, and optional Applicant Tracking Systems (ATS) through standardized APIs.

This system establishes a clear boundary: all AI processing, resume parsing, and accessibility services occur within the system, while data storage and optional integrations rely on secure third-party infrastructure. The context diagram ensures stakeholders understand how external entities interact with the system and which responsibilities lie inside versus outside the system boundary.

Product Boundary and Limitations:

The first release of the system does not include automated interview analysis, automated job-matching recommendations, or direct integration with external company HR systems beyond optional ATS API connections. These capabilities may be considered in future versions but are explicitly out of scope for Version 1.0. By defining these boundaries, stakeholders have a clear understanding of the system’s scope and intended functionality.

Non-Goals:

The following items are explicitly not part of the system’s goals for Version 1.0:

- Automated video-interview evaluation or speech analysis

- Predictive performance scoring beyond skill inference
- Automated job-fit ranking across multiple open roles
- Integration with payroll or onboarding systems
- Behavioral or personality profiling of applicants
  These non-goals ensure the system remains focused on accessibility, fairness, and transparent resume-based evaluation.

2.2 Product Features

At a high level, the Inclusive and Accessible Hiring Technology will deliver the following major features:

- **Resume Parsing and Skill Inference:** AI-powered parsing of resumes in multiple formats, extracting skills and experience, and allowing candidate review and correction.
- **Voluntary Disclosure and Accommodation Handling:** Secure mechanism for candidates to disclose accommodations while maintaining privacy and separation from evaluation data.
- **Role-Based Interfaces:** Separate, accessible interfaces tailored for applicants, recruiters, hiring managers, and HR teams, compliant with WCAG 2.1 AA standards.
- **Fairness, Transparency, and Explainability:** Explainable AI logs, decision summaries, and audit capabilities to foster trust and compliance.
- **Accessibility and Compliance:** Full adherence to accessibility standards (WCAG 2.1 AA, ADA, AODA) and data privacy laws (GDPR, PIPEDA).
- **HR/DEI Analytics:** Dashboards for diversity metrics, audit logs, and compliance reporting.
- **External System Integration:** Optional, standards-based API connections allowing the system to export candidate data, import job postings, and synchronize application statuses with external Applicant Tracking Systems (ATS). These integrations maintain system independence while supporting streamlined workflows for organizations that use existing recruitment platforms.
- **Notifications and Communications:** Accessible, role-specific notification features such as email alerts and in-platform messages for applicants, recruiters, and HR/DEI staff. These notifications follow accessibility guidelines and ensure users receive timely updates on application progress, review tasks, and audit events.

These features will work together to ensure an inclusive, transparent, and fair hiring process that benefits all stakeholders.

Table 2-1: Traceability Matrix

Feature ID	Product Feature	Related RD Requirement(s)
PF-1	Resume Parsing and Skill Inference	REQ-1.1, REQ-1.2, REQ-1.3, REQ-1.4
PF-2	Voluntary Disclosure and Accommodation Handling	REQ-2.1, REQ-2.2, REQ-2.3, REQ-2.4, REQ-2.5
PF-3	Role-Based Interfaces	REQ-3.1, REQ-3.2, REQ-3.3, REQ-3.4, REQ-3.5
PF-4	Fairness, Transparency and Explainability	REQ-4.1, REQ-4.2, REQ-4.3, REQ-4.4, REQ-4.5
PF-5	Accessibility and Compliance	REQ-5.1, REQ-5.2, REQ-5.3, REQ-5.4, REQ-5.5
PF-6	HR/DEI Analytics	REQ-6.1, REQ-6.2, REQ-6.3, REQ-6.4
PF-7	External System Integration	REQ-4.3.1, REQ-4.3.2, REQ-4.3.3, REQ-4.3.4, REQ-4.3.5
PF-8	Notifications and Communications	REQ-4.4.1, REQ-4.4.2, REQ-4.4.3, REQ-4.4.4, REQ-4.4.5

Explanation for Traceability Matrix:

The traceability matrix ensures that every functional requirement is clearly linked to a specific system feature, providing a structured and verifiable connection between requirements and design decisions. This alignment is critical for demonstrating completeness in the RD and ensures the development team can verify that all stated requirements will be implemented and tested against specific system features.

For instance:

- **REQ-1.1 to REQ-1.4** are mapped to “Resume Parsing and Skill Inference,” confirming that parsing diverse formats and inferring skills are core functionalities.

Rationale: These requirements ensure the system can accurately interpret diverse resume formats and extract relevant skills without disadvantaging applicants who use non-traditional layouts, assistive technologies, or have career gaps. By supporting correction and validation by the applicant, the system increases data accuracy while maintaining transparency and user control. This directly supports the project's goals of fairness, inclusivity, and accessible evaluation.

- **REQ-2.1 to REQ-2.5** correspond to "Voluntary Disclosure and Accommodation Handling," which secures voluntary disclosure while maintaining privacy and compliance.

  Rationale: These requirements are necessary to allow candidates to disclose accommodation needs securely while ensuring the information does not influence the evaluation process. The rationale is to uphold privacy laws (GDPR, PIPEDA) and reduce bias by separating personal accommodation data from screening workflows. This aligns with the project objective of creating an inclusive and equitable hiring experience for applicants with diverse needs.

- **REQ-3.1 to REQ-3.5** match "User Role-Based Interfaces" to ensure that interface designs meet accessibility standards and provide role-specific functions.

  Rationale: These requirements ensure each user class—applicants, recruiters, hiring managers, and HR/DEI staff—receives an interface tailored to their responsibilities and accessibility needs. By designing role-specific, WCAG-compliant interfaces, the system improves usability, reduces cognitive load, and ensures equal access for all users, including those with disabilities. This supports the high-level design goal of creating an accessible and efficient multi-stakeholder system.

- **REQ-4.1 to REQ-4.5** are tied to "Fairness, Transparency, and Explainability," ensuring the system provides clear reasoning for AI decisions and supports manual overrides.

  Rationale: These requirements ensure that AI-driven decisions are understandable, traceable, and open to human oversight. Including decision explanations, audit logs, and manual override capabilities supports regulatory compliance and builds trust among users. The rationale is to mitigate algorithmic bias, provide accountability, and meet emerging standards in ethical AI governance—key objectives identified in the challenge statement.

- **REQ-5.1 to REQ-5.5** relate to "Accessibility and Compliance," reinforcing adherence to accessibility standards and data privacy requirements.

  Rationale: These requirements guarantee that the system adheres to WCAG 2.1 AA and privacy regulations such as GDPR and PIPEDA. The rationale is to ensure all applicants, regardless of ability or device, can use the system effectively while maintaining secure data

handling practices. Compliance with these standards is essential for legal, ethical, and operational reasons, and ensures the platform is trustworthy for organizations and applicants.

By documenting these relationships, the matrix supports both design clarity and verification/testing processes. It also enables stakeholders to track progress and ensure the final solution aligns with the intended inclusive hiring objectives.

2.3 User Classes and Characteristics

The system is intended to serve the following user classes:

- **High-Level User Needs Overview**

  1- All primary user groups require an accessible, reliable, and transparent experience throughout the hiring process. Applicants need clear communication, control over their data, and equitable evaluation. Recruiters and hiring managers need accurate, explainable insights that reduce workload without sacrificing fairness. HR and DEI teams require auditability, compliance reporting, and visibility into system-wide trends. These shared needs inform the design principles and features described throughout this section.

- **Applicants**

  1- Characteristics: Individuals applying for jobs with varying degrees of technical proficiency. May include people with disabilities, non-linear career paths, or those who require accommodations.

  2- Needs: Accessibility-compliant interface, ability to track application status, control disclosure of personal information, and preview/edit parsed resume data.

- **Recruiters**

  1- Characteristics: HR professionals responsible for initial candidate screening and shortlisting.

  2- Needs: Clear candidate summaries, skill matches, AI explainability, ability to override AI decisions, and access to diversity and compliance data.

- **Hiring Managers**

1- Characteristics: Decision-makers evaluating shortlisted candidates for specific roles.

2- Needs: Access to candidate portfolios, cover letters, skill profiles, and explanations for AI-based recommendations.

- **Human Resources (HR) / Diversity, Equity, and Inclusion (DEI) Teams**

1- Characteristics: Teams responsible for ensuring compliance and fairness in the recruitment process.

2- Needs: Detailed audit logs, diversity metrics dashboards, anonymized reports, and compliance tracking tools.

2.4 Operating Environment

The system will operate in a modern web-based environment and must function reliably across multiple platforms and devices. The operating environment includes:

- **Hardware**: Desktop, laptop, tablet, and smartphone devices.
- **Operating Systems**: Windows, macOS, Linux, Android, and iOS.
- **Browsers**: Latest stable versions of Chrome, Edge, Firefox, and Safari.
- **Backend**: Cloud-based server infrastructure (e.g., AWS, Azure) with API endpoints.
- **Databases**: Relational database management systems for storing candidate data, parsed resumes, audit logs, and accommodation disclosures.
- **Security**: HTTPS communication, encryption of data at rest and in transit, secure authentication mechanisms, and compliance with GDPR/PIPEDA.

2.5 Design and Implementation Constraints

The following constraints will influence the design and implementation of the system:

- **Accessibility Standards**: Must comply with WCAG 2.1 AA standards, including screen reader compatibility, high contrast mode, keyboard navigation, and voice input.

- **Privacy and Security**: Must adhere to GDPR, PIPEDA, and ADA regulations, ensuring separate storage of accommodation disclosures, encryption, and strict access control.

- **Integration Constraints**: Integration with legacy ATS platforms will require standardized APIs and adherence to different data formats.

- **Performance Constraints**: The system must load interfaces within 1 second and support at least 50,000 concurrent users without performance degradation.

- **Development Constraints**: Use of existing AI and NLP models where possible to reduce development time and improve accuracy.

2.6 Assumptions and dependencies

The following assumptions and dependencies have been identified for the project:

- **Assumptions**:
    - Candidates will be willing to voluntarily disclose accommodation needs.
    - Recruiters and hiring managers will adopt and trust AI-supported screening tools if explanations are transparent.
    - The underlying NLP and AI models will be sufficiently trained on diverse datasets to reduce bias.
    - The system will be hosted on a secure, scalable cloud platform.
- **Dependencies**:
    - Availability of reliable AI/NLP models and frameworks for resume parsing.
    - Access to applicant tracking systems' APIs for integration.
    - Cloud hosting infrastructure with sufficient performance and availability guarantees.
    - Continued compliance with evolving accessibility and privacy regulations.

3.0 System Features

3.1 Resume Parsing and Skill Inference

3.1.1 Description and Priority

The system must parse resumes of various formats and infer both core and related skills using AI-driven algorithms. It supports parsing of resumes generated via assistive technologies and allows candidates to preview and correct the parsed data before submission.

Priority: **High**

3.1.2 Functional Requirements

- **REQ-1.1:** The system shall support PDF, DOCX, plain text, and screen-reader compatible resume formats at launch.
- **REQ-1.2:** The system shall accurately extract core skills and infer related competencies using an ontology-backed AI model.
- **REQ-1.3:** The system shall allow candidates to manually edit or supplement skill data before submission.
- **REQ-1.4:** The parser shall support resumes with non-standard layouts or embedded media by offering manual form-filling fallback.

3.1.3 Use Cases

UC-1-1: Uploading a Resume

A user uploads their resume to the system.

Actors

User

Secondary Use Cases

N/A

Preconditions

The user must have an account and be logged in.

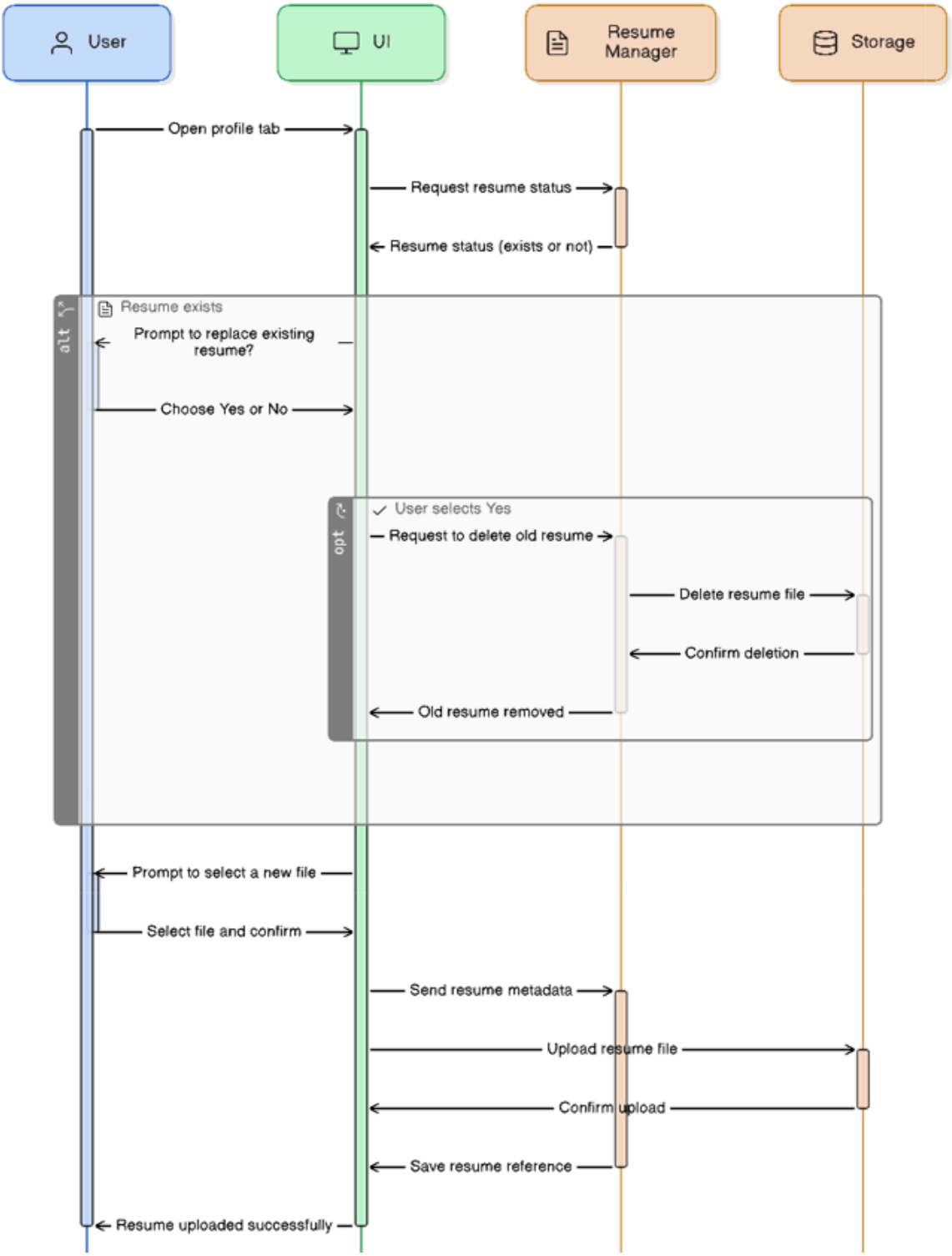
Postconditions

The system will use the data harvested from the resume to employers when the user applies for a job.

Main Flow

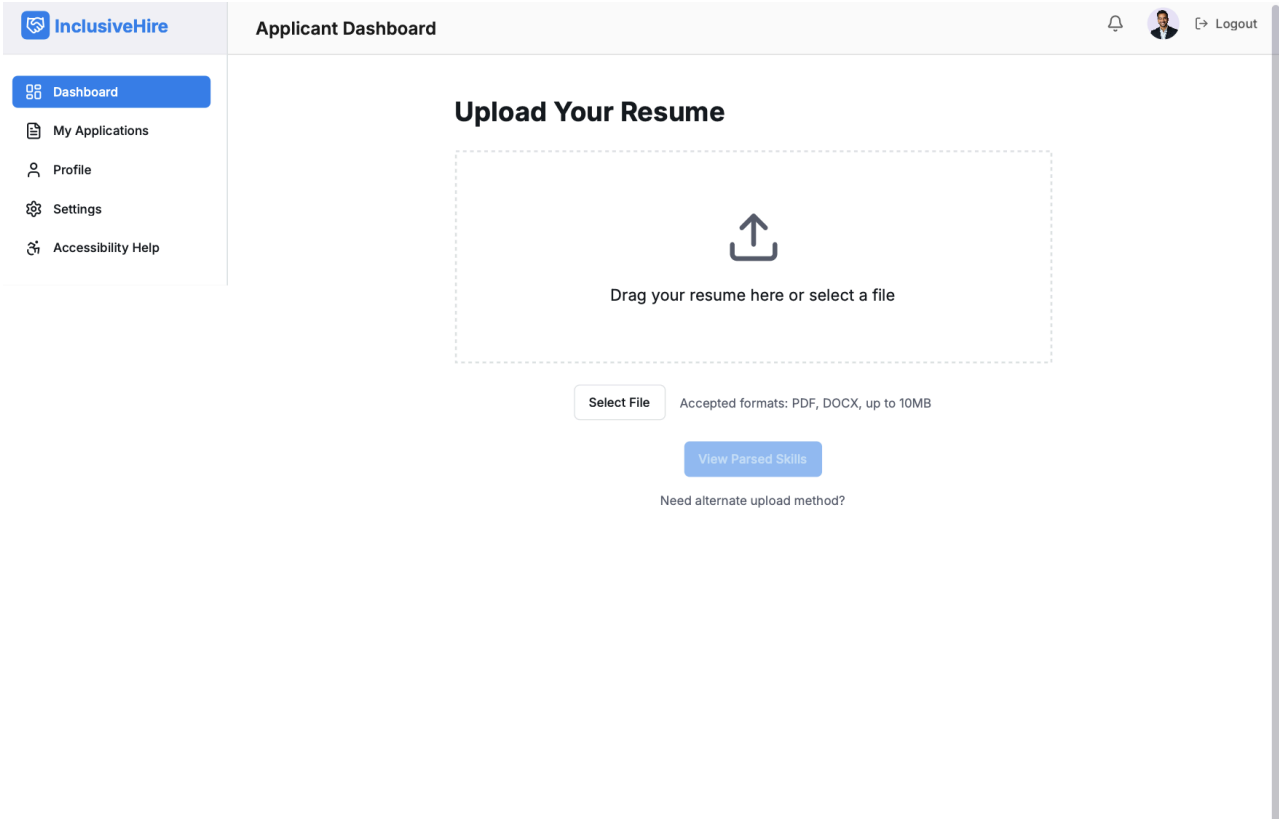
1. The user navigates to their “profile” tab.
2. The user clicks the “Upload Resume” button.
3. If the user already has a resume uploaded:
    3.1. The user is prompted if they want to remove their old resume.
    3.2. If they click yes:
        3.2.1. The old resume is removed from the profile.
    3.3. If they click no:
        3.3.1. The old resume stays as the current resume
        3.3.2. The prompt disappears.
4. The user is prompted to select a new file.
5. The user confirms the addition.

UC-1-1: Sequence Diagram



UC 1-1: User Story

As a registered user, I want to upload my resume to the system, so that the system can extract my information and make it available to employers when I apply for jobs.



3.2 Voluntary Disclosure and Accommodation Handling

3.2.1 Description and Priority

The system must allow applicants to voluntarily disclose accommodation needs through a secure, separate process not used in candidate matching.

Priority: **High**

3.2.2 Functional Requirements

- **REQ-2.1:** The system shall provide a separate interface or input method for disclosing accommodation needs.
- **REQ-2.2:** Disclosed data shall be encrypted and stored separately from application data.

- **REQ-2.3:** The system shall prevent recruiters from searching for candidates based on disclosed accommodations before screening.
- **REQ-2.4:** The system shall support all accessibility-related accommodation types.
- **REQ-2.5:** Consent must be obtained before using disclosed data for auditing or reporting purposes.

3.2.3 Use Cases

UC- 2-1: Submit Voluntary Disclosure

A user privately submits accommodation needs through a separate, secure interface.

Actors
User

Secondary Use Cases
N/A

Preconditions
The user has an account in the system; the user is authenticated and logged in.

Postconditions
Disclosure is encrypted and stored separately; consent captured; audit entry created; not visible/searchable to recruiters before screening.

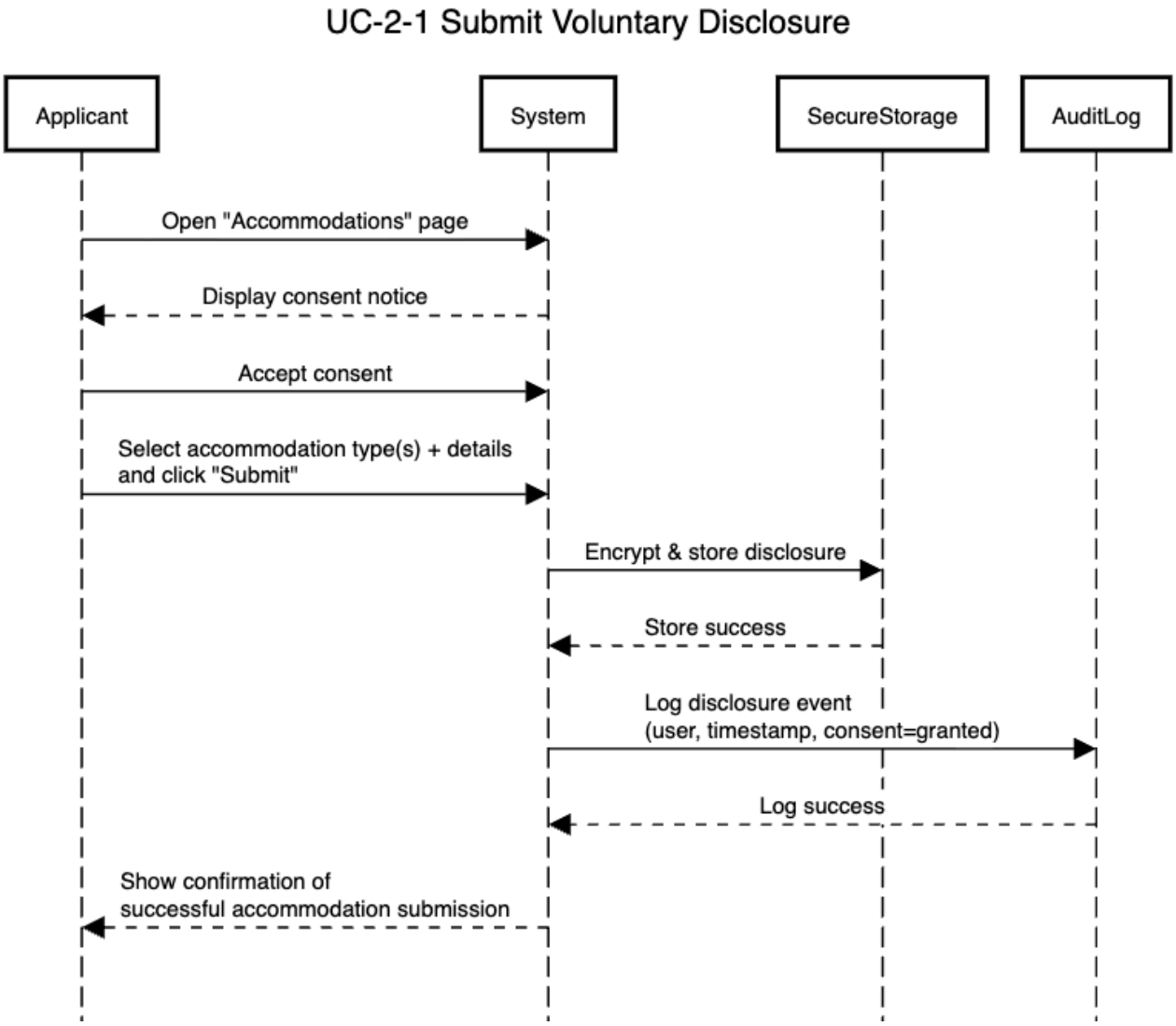
Main Flow

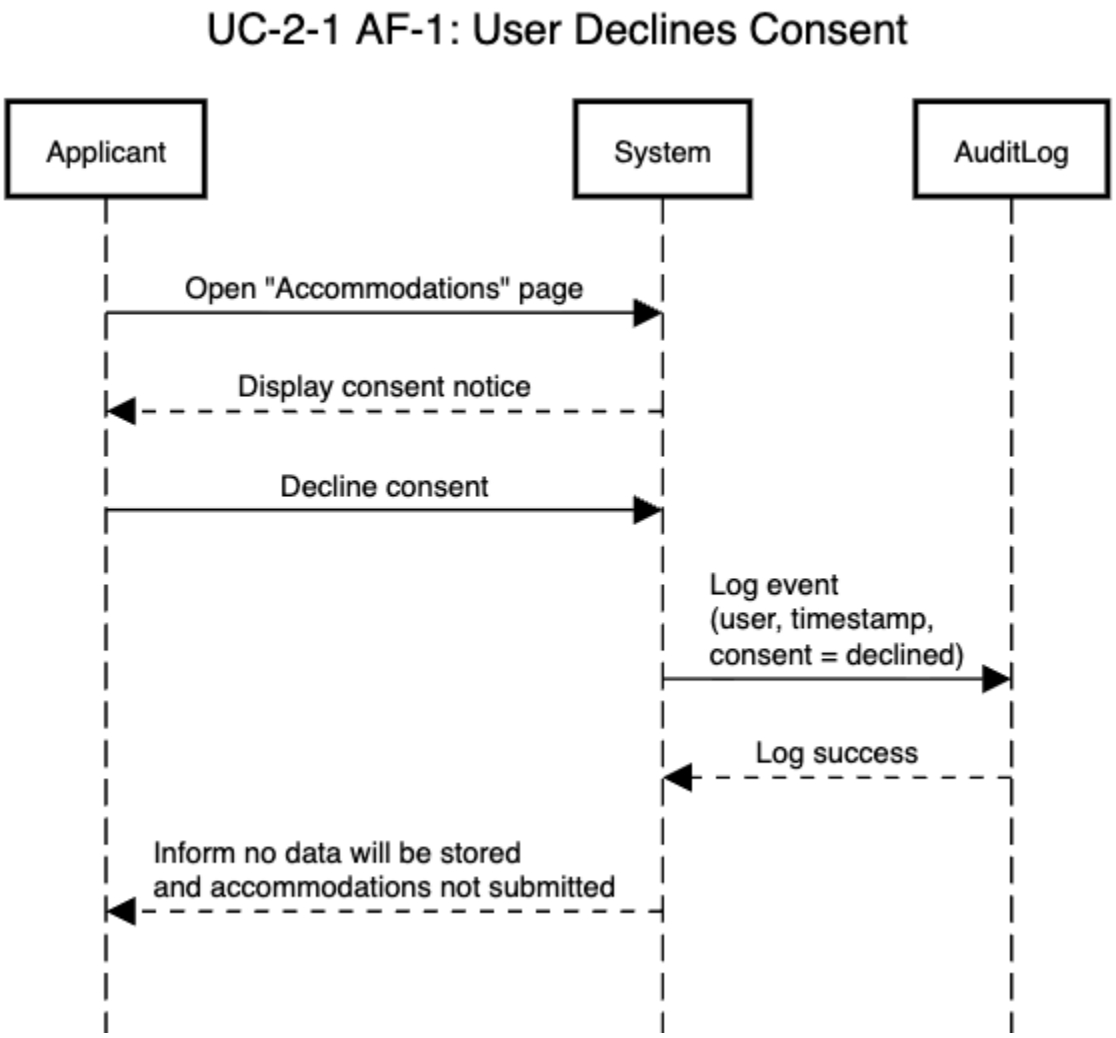
1. User opens **Accommodations** (separate from application).
2. The system shows a consent notice explaining how accommodation information will be stored and used.
3. The user selects accommodation type(s) and submits.
4. The system encrypts and stores disclosure in segregated storage.
5. System writes audit log; recruiter views remain unchanged.

Alternate Flow

1. Consent declined → nothing saved; audit records “consent declined.”

UC- 2-1: Sequence Diagram





UC 3-1: User Story

As a user with accessibility needs, I want to privately submit my accommodation requirements through a secure interface, so that I can receive the necessary support during the application and interview process without revealing sensitive information to recruiters prematurely.

3.3 User Role-Based Interfaces

3.3.1 Description and Priority

Distinct interfaces must be designed for each user group: applicants, recruiters, hiring managers, and HR, supporting role-specific functions.

Priority: **High**

3.3.2 Functional Requirements

- **REQ-3.1:** Applicants shall be able to track their application status, edit parsed data, and control what personal information is disclosed.

  Rationale: This requirement ensures applicants retain agency over their data and visibility into the hiring process, reducing anxiety, correcting parsing errors, and supporting privacy and trust in the system.

- **REQ-3.2:** Recruiters shall receive shortlists with plain-language summaries of skills and the ability to override AI suggestions.

  Rationale: This requirement ensures recruiters can work efficiently with AI support while retaining final human judgment, improving usability, reducing misinterpretation of AI outputs, and preventing over-reliance on automated decisions.

- **REQ-3.3:** Hiring Managers shall view extended candidate materials such as portfolios and cover letters.

  Rationale: This requirement ensures hiring managers can make well-informed decisions by reviewing richer evidence of candidate suitability beyond resumes and AI scores, supporting fair and context-aware evaluations.

- **REQ-3.4:** HR shall access tamper-evident audit logs and track diversity and inclusion metrics.

  Rationale: This requirement ensures HR can monitor system behavior, detect inappropriate actions or bias, and demonstrate compliance with internal policies and external regulations through verifiable, non-repudiable records.

- **REQ-3.5:** The system shall ensure all interfaces meet WCAG 2.1 AA standards or higher.

  Rationale: This requirement ensures the platform is accessible to users with diverse abilities, reducing barriers to participation and aligning the system with recognized accessibility standards and legal expectations.

3.3.3 Use Cases

UC-3-1: View Application Status

A user views their current job-application status through the applicant interface

Actors

Applicant

Secondary Use Cases

UC-1-1 Uploading a Resume.

Preconditions

- The applicant is logged into their account.
- At least one job application has been submitted.

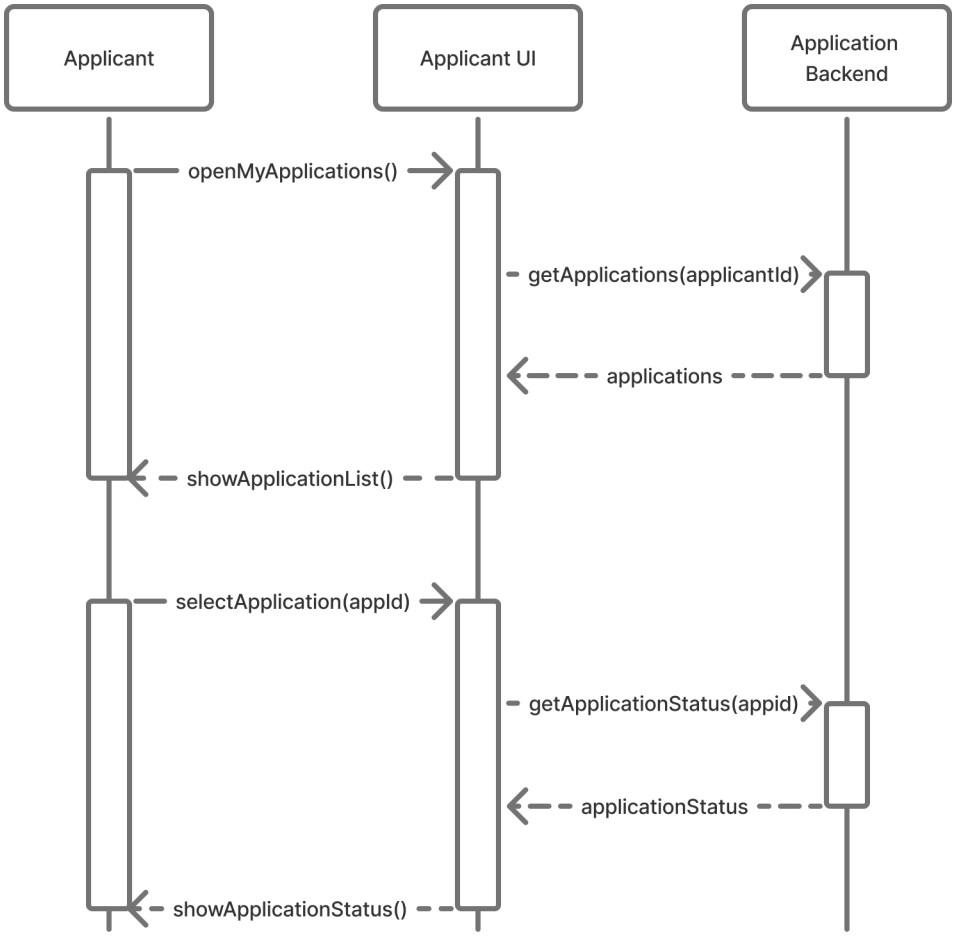
Postconditions

- The applicant can see the most recent status of each submitted application (e.g., Received, Under Review, Shortlisted, Declined).

Main Flow

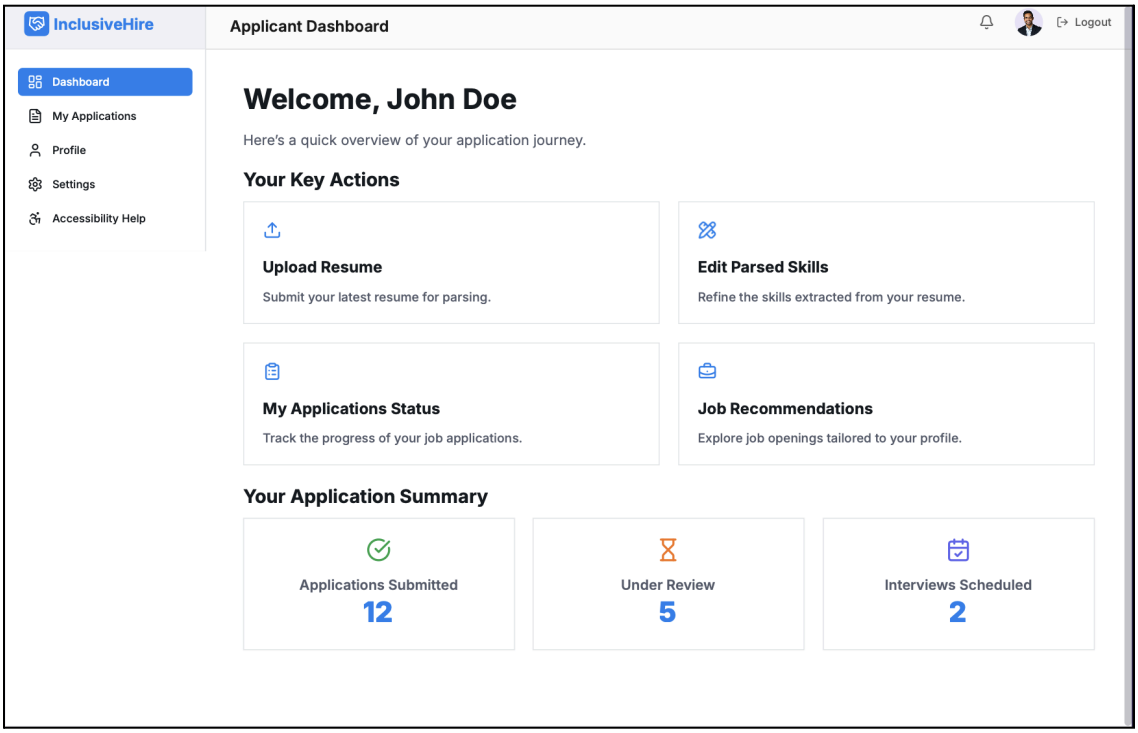
1. The applicant selects the “My Applications” tab from the navigation bar.
2. The system displays a list of all submitted applications.
3. The applicant selects an application from the list.
4. The system displays the current application status in plain language and the date of the last update.
5. The applicant may choose to return to the list or navigate to edit their profile information.

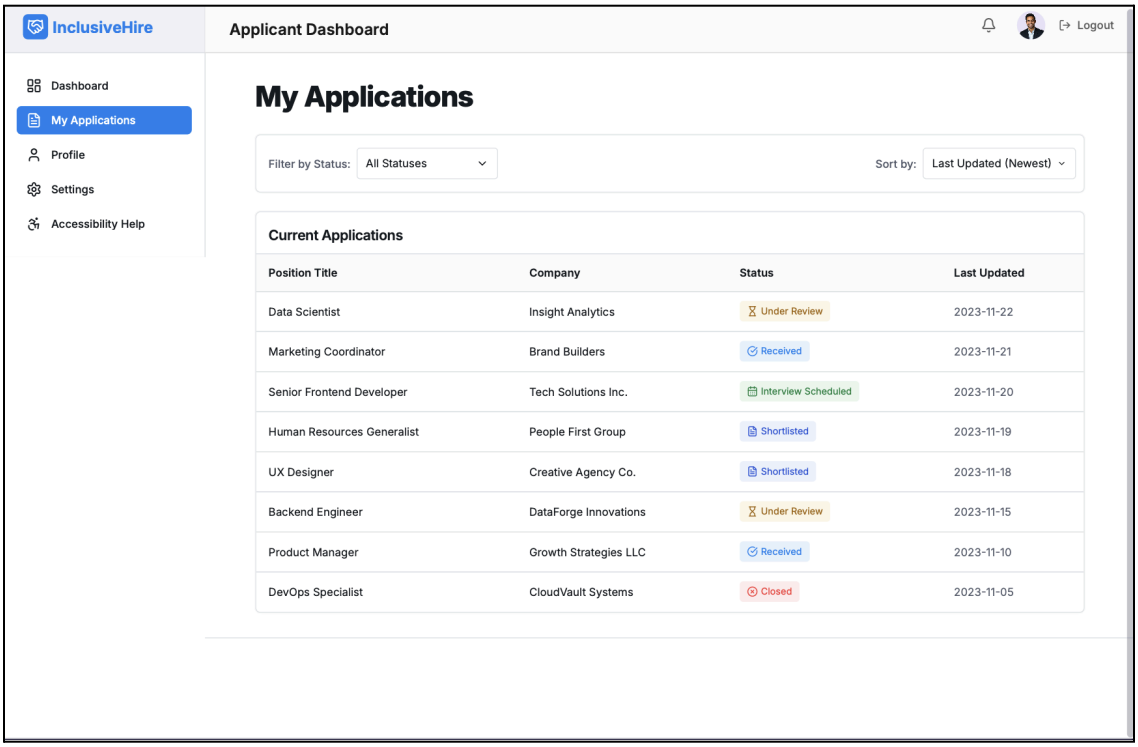
UC-3-1: Sequence Diagram



UC-3-1: User Story

As an applicant, I want to view the status of my submitted job applications so that I can stay informed and understand where I am in the hiring process.


InclusiveHire
Applicant Dashboard
Logout
Dashboard
My Applications
Profile
Settings
Accessibility Help
Welcome, John Doe
Here's a quick overview of your application journey.
Your Key Actions
Upload Resume
Submit your latest resume for parsing.
Edit Parsed Skills
Refine the skills extracted from your resume.
My Applications Status
Track the progress of your job applications.
Job Recommendations
Explore job openings tailored to your profile.
Your Application Summary
Applications Submitted
12
Under Review
5
Interviews Scheduled
2


InclusiveHire
Applicant Dashboard
Logout
Dashboard
My Applications
Profile
Settings
Accessibility Help
My Applications
Filter by Status:
All Statuses
Sort by:
Last Updated (Newest)
Current Applications
Position Title
Company
Status
Last Updated
Data Scientist
Insight Analytics
Under Review
2023-11-22
Marketing Coordinator
Brand Builders
Received
2023-11-21
Senior Frontend Developer
Tech Solutions Inc.
Interview Scheduled
2023-11-20
Human Resources Generalist
People First Group
Shortlisted
2023-11-19
UX Designer
Creative Agency Co.
Shortlisted
2023-11-18
Backend Engineer
DataForge Innovations
Under Review
2023-11-15
Product Manager
Growth Strategies LLC
Received
2023-11-10
DevOps Specialist
CloudVault Systems
Closed
2023-11-05

The mockup illustrates the applicant's "My Applications" interface, which displays all submitted applications along with their current system-generated statuses. This supports transparency and aligns with the user story by allowing applicants to track their hiring progress.

UC-3-2: Hiring Manager Views Candidate Materials

A hiring manager views extended candidate materials for a posting.

Actors

Hiring Manager

Secondary Use Cases

N/A

Preconditions

- The hiring manager is logged in.
- Candidates are associated with the selected job posting

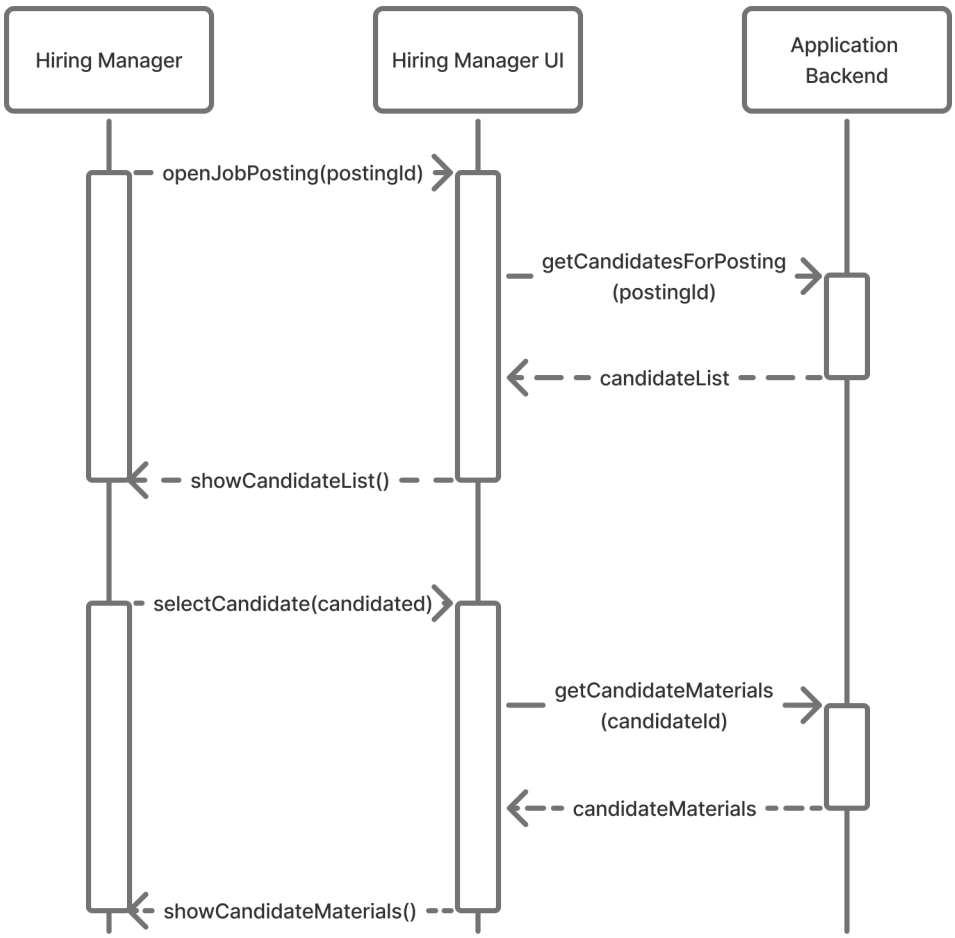
Postconditions

- Extended materials for the selected candidate are displayed.

Main Flow

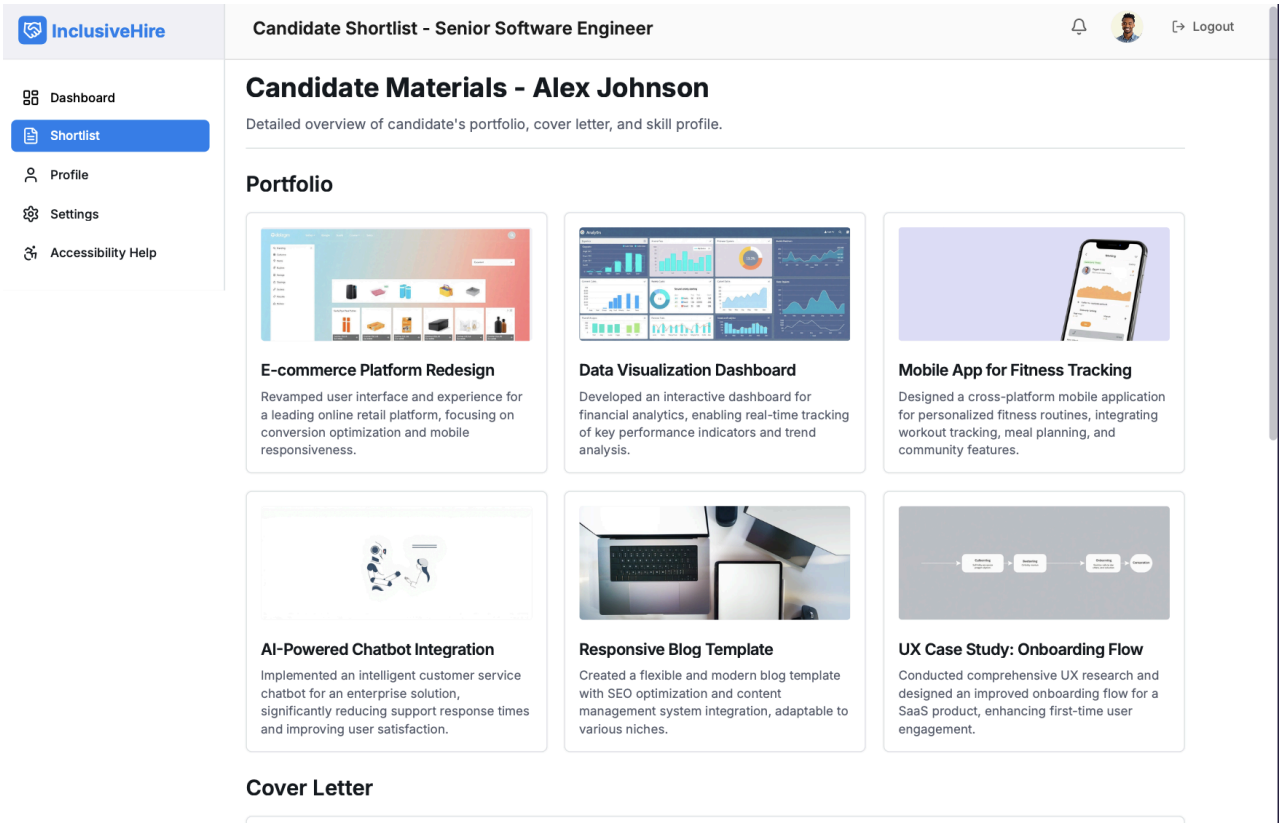
1. The hiring manager opens the job posting page.
2. The system shows the candidate list for the posting.
3. The hiring manager selects a candidate.
4. The system displays the candidate's portfolio, cover letter, and skill profile.
5. The hiring manager returns to the list or selects another candidate.

UC-3-2: Sequence Diagram



UC 3-2: User Story

As a hiring manager, I want to view extended materials for candidates associated with a job posting, So that I can evaluate their qualifications, portfolio, and skills before making a hiring decision.



UC-3-3: HR Accesses Audit Logs

An HR user views the system's audit logs to review recorded system actions and verify compliance.

Actors

HR / DEI Team Member.

Secondary Use Cases

N/A

Preconditions

- The HR user is logged into the system.
- The system has existing audit logs of user or AI actions.

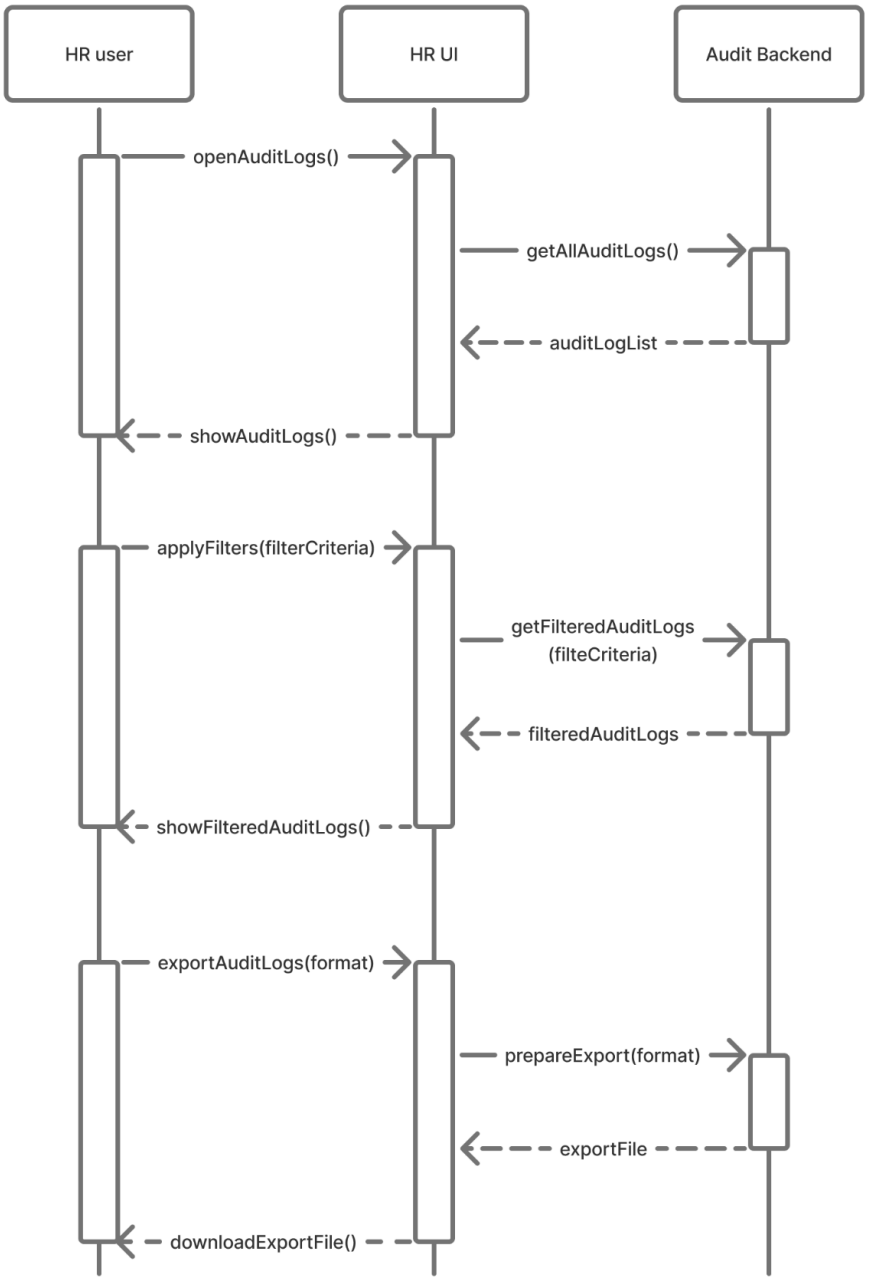
Postconditions

- The HR user has viewed audit log entries; no data has been altered.

Main Flow

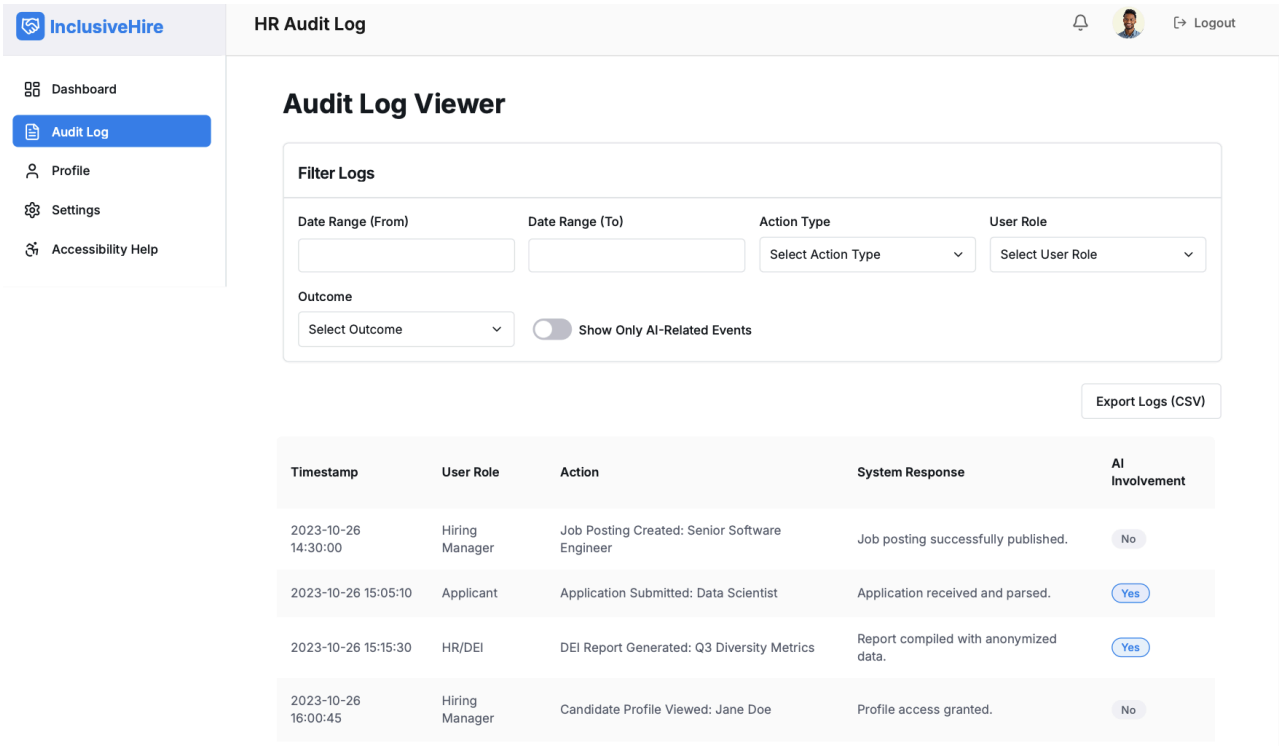
1. The HR user navigates to the “Audit Logs” section in the HR interface.
2. The system retrieves a list of logged system actions (e.g., AI decisions, recruiter overrides, data access).
3. The HR user filters logs by date range or user role.
4. The system displays the filtered audit records.
5. The HR user may download or export the log for reporting.

UC-3-3: Sequence Diagram



UC 3-3: User Story

As an HR or DEI team member, I want to access the system's audit logs, so that I can review recorded system actions and verify compliance without altering any data.



3.4 Fairness, Transparency, and Explainability

3.4.1 Description and Priority

The system must include features that make all AI decisions explainable and auditable for compliance and candidate confidence.

Priority: **High**

3.4.2 Functional Requirements

- **REQ-4.1:** All AI decisions must be logged with reason codes and feature importances.
- **Rationale:** This requirement ensures transparency by capturing the factors that influenced each AI decision, supporting accountability, explainability, and compliance with audit and fairness standards.

- **REQ-4.2:** The system shall provide one-sentence explanations for each candidate's shortlist inclusion.
- **Rationale:** This requirement ensures that users receive a clear and concise summary of why a candidate was selected, improving decision clarity and enabling quicker recruiter review.
- **REQ-4.3**: Recruiters and candidates shall be able to view AI decision logs in plain English.
- **Rationale:** This requirement ensures accessibility and usability for all users by presenting AI outputs in simple, understandable language rather than technical jargon, reducing confusion and supporting informed decision-making.
- **REQ-4.4:** The system shall allow manual correction of AI decisions by candidates and overrides by recruiters.
- **Rationale:** This requirement ensures fairness and human oversight by allowing users to identify inaccurate data and enabling recruiters to override AI outcomes when necessary, maintaining trust and preventing incorrect automation-driven decisions.
- **REQ-4.5:** Monthly audit reports shall be automatically generated for HR.
- **Rationale:** This requirement ensures that HR has consistent visibility into AI decision patterns, overrides, and corrections, supporting compliance monitoring, bias detection, and continuous system improvement.

3.4.3 Use Cases

UC-4-1: View AI Decision Explanation

A recruiter views the explanation for a candidate's shortlist inclusion.

Actors

Recruiter

Secondary Use Cases

N/A

Preconditions

- The AI system has produced a shortlist of candidates.
- Explanation data (reason codes, feature importances) has been stored in the decision log.

Postconditions

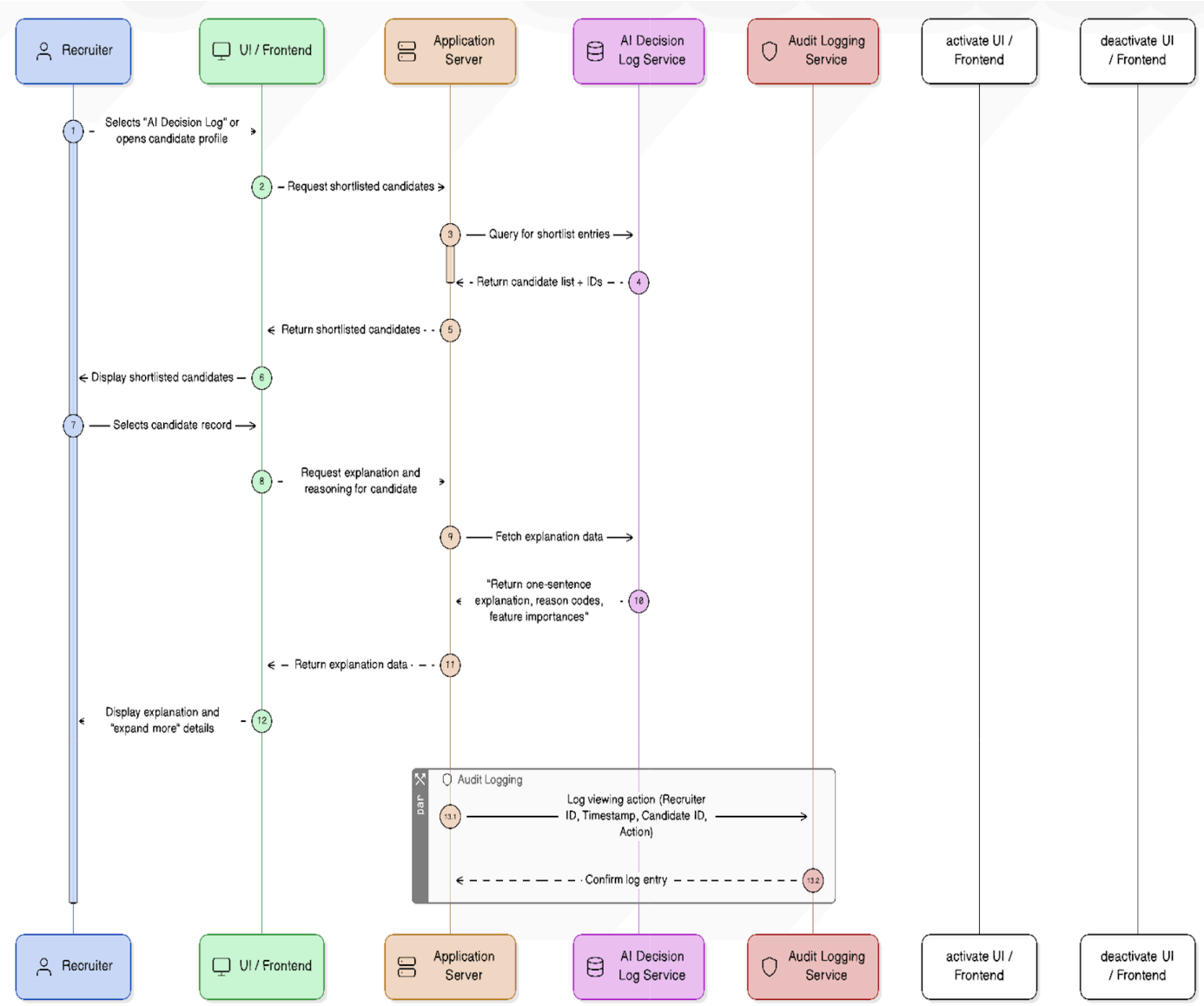
- The recruiter successfully views the one-sentence explanation and expanded reasoning.

- The viewing event is logged for audit and compliance purposes.

Main Flow

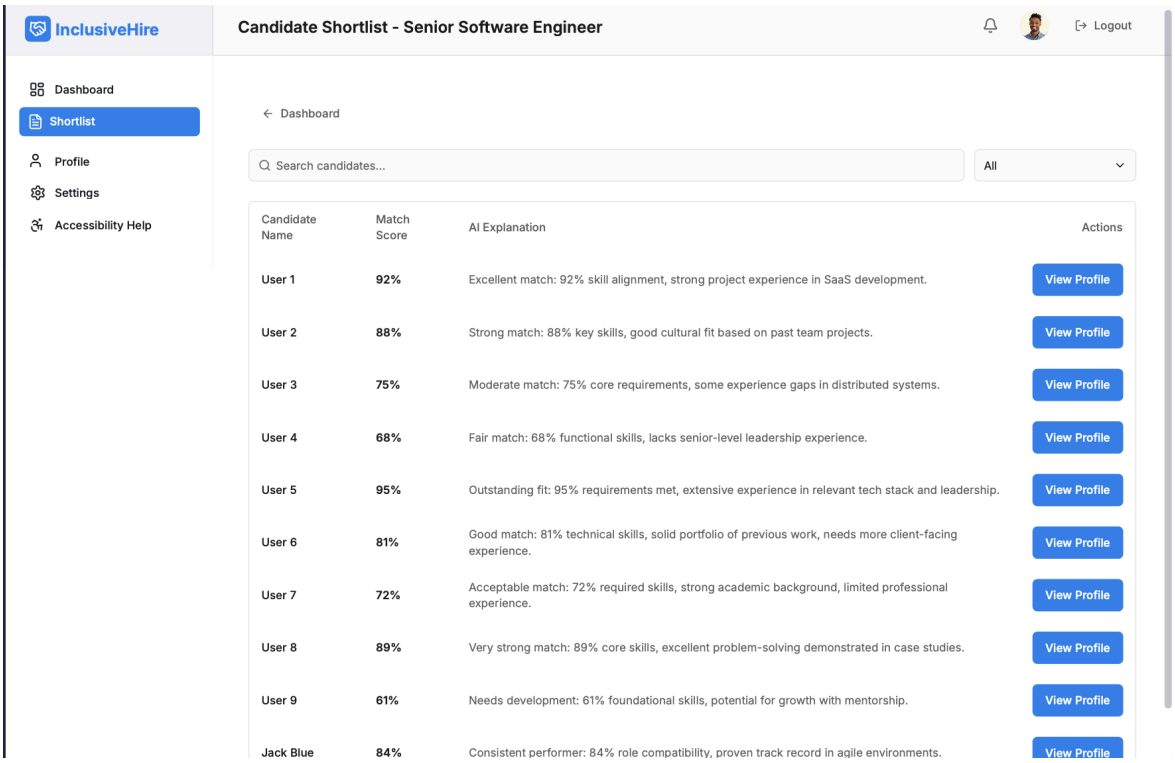
1. The recruiter navigates to the “AI Decision Log” or candidate view.
2. The system displays the list of shortlisted candidates.
3. The recruiter selects the relevant record.
4. The system displays the one-sentence explanation of why the candidate was included.
5. The user may expand the explanation to view key features and reason codes.
6. The system logs the viewing event for audit purposes.

UC-4-1: Sequence Diagram



UC-4-1: User Story

As a Recruiter, I want to view a clear explanation of why a candidate was included in the AI-generated shortlist, so that I can understand the reasoning behind the recommendation and make an informed hiring decision.



UC-4-2: Submit Correction or Feedback on AI Decision

A candidate submits feedback or a correction request after they have received a status update

Actors

Candidate

Secondary Use Cases

UC-3-1 (View Application Status)

Preconditions

- The candidate can view their application decisions.
- The system accepts feedback submissions tied to a logged decision.

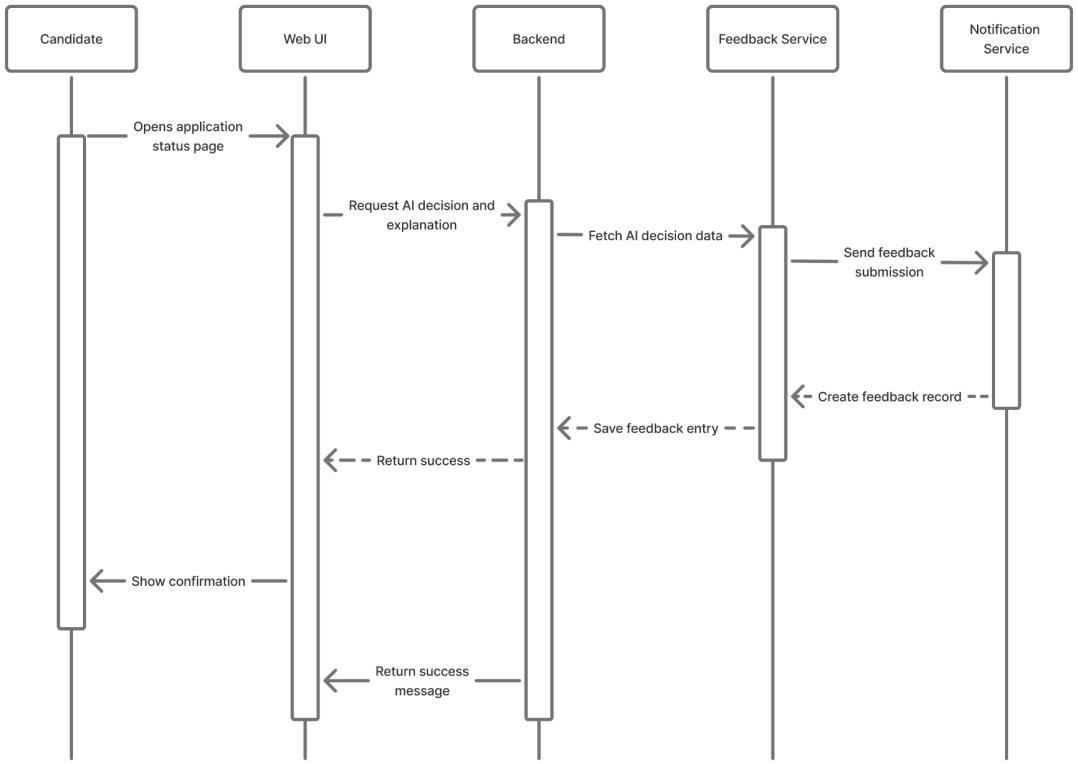
Postconditions

- A correction or feedback record is created and stored for recruiter review.
- The AI decision itself remains unchanged until a recruiter review occurs.

Main Flow

1. The candidate views their application status.
2. The status says “not selected”.
3. The candidate selects “Request Review” or “Submit Correction.”
4. The system displays a form for clarifying missing or misinterpreted information.
5. The candidate provides details and submits the correction.
6. The system stores the request and notifies the assigned recruiter or HR for validation.

UC-4-2: Sequence Diagram



UC-4-3: Override AI Decision

A recruiter manually overrides an AI shortlist decision after reviewing explanations or candidate feedback.

Actors

Recruiter

Secondary Use Cases

UC-4-1 (View AI Decision Explanation)

UC-4-2 (Submit Correction or Feedback on AI Decision)

Preconditions

- The recruiter has appropriate permissions.
- The AI decision and explanation are visible in the recruiter interface.

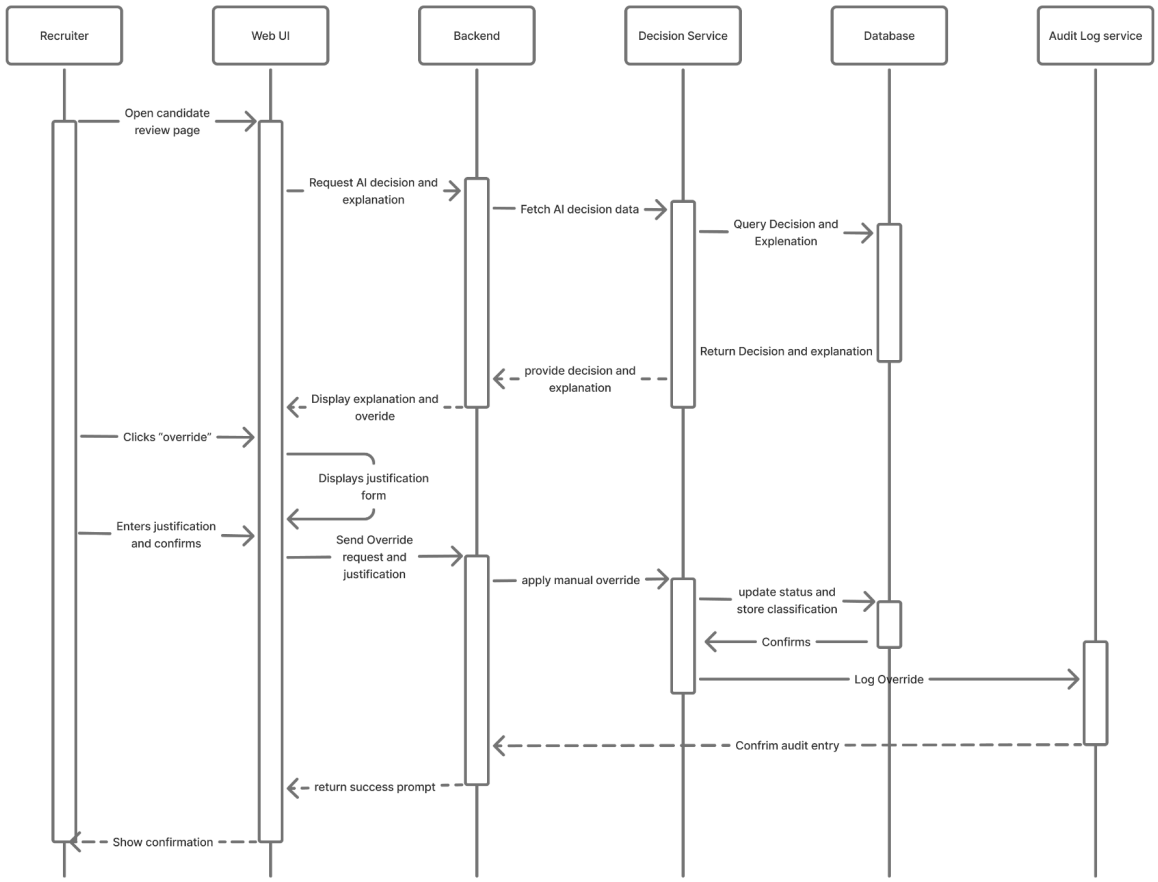
Postconditions

- The recruiter's override is applied, recorded with justification, and included in the next audit report.

Main Flow

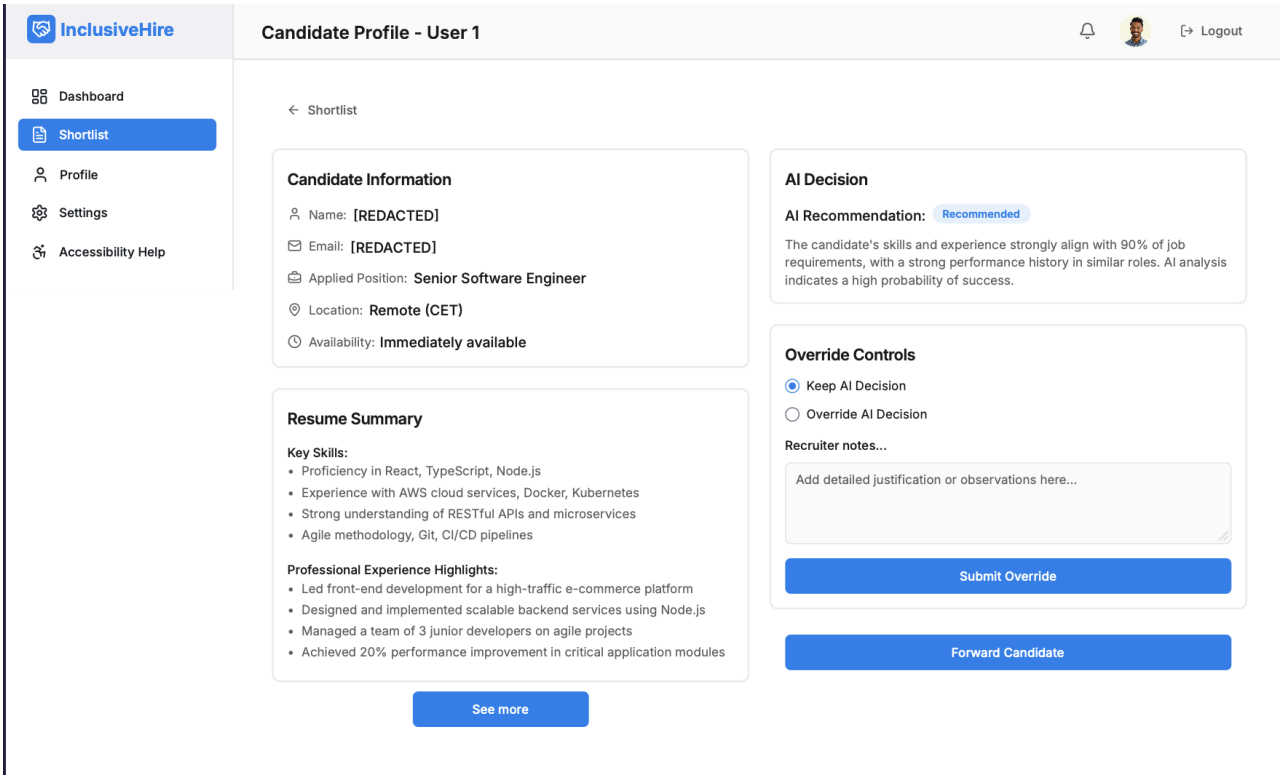
1. The recruiter reviews the candidate's AI decision explanation.
2. The recruiter selects "Override Decision."
3. The system prompts for a justification.
4. The recruiter provides a reason and confirms.
5. The system updates the candidate's status and logs the override with timestamp, user ID, and justification.
6. The system includes this record in the monthly audit summary.

UC-4-3: Sequence Diagram



UC 4-3: User Story

As a recruiter with appropriate permissions, I want to manually override an AI-generated shortlist decision after reviewing the explanation and candidate feedback, so that I can ensure fair, context-aware hiring decisions when the AI output is incomplete, incorrect, or requires human judgment.



3.5 Accessibility and Compliance

3.5.1 Description and Priority

All components of the system must adhere to strict accessibility and data compliance regulations.

Priority: **High**

3.5.2 Functional Requirements

- **REQ-5.1:** The system must comply with WCAG 2.1 AA and support screen readers, keyboard-only navigation, high contrast mode, and voice input.

  Rationale: This rule is crucial since it ensures that everyone, including those with disabilities, can utilize the system. The system becomes more user-friendly and inclusive by enabling features like voice input, keyboard navigation, high contrast, and screen readers. Additionally, it assists us in meeting accessibility requirements so that all candidates may utilize the system equitably.

- **REQ-5.2:** Accessibility compliance audits of the user interfaces shall be conducted every four (4) weeks by the development or QA team using recognized accessibility evaluation tools (e.g., WAVE, Axe, or Lighthouse) and manual testing against WCAG 2.1 AA criteria to verify continued conformance and identify regressions introduced by updates.

Rationale: The team can identify new problems that may arise following updates or changes by conducting accessibility audits every four weeks. This is crucial because accessibility features might be unintentionally broken by even minor UI changes. Frequent testing guarantees that we don't fall below the necessary WCAG 2.1 AA standards and keeps the system constantly accessible.

- **REQ-5.3:** The system must retain data for 5 years and meet global compliance standards (e.g., GDPR, PIPEDA, AODA).

  Rationale: Since many HR and legal procedures call for long-term record preservation, data must be retained for five years. Respecting international privacy regulations such as GDPR, PIPEDA, and AODA contributes to the system's compliance and reliability. Additionally, it guarantees that the product can be used in various areas without violating any regulations.

- **REQ-5.4:** Candidates shall have the right to delete their data ("right to be forgotten").

  Rationale: Giving applicants the option to remove their data increases their control over their personal information and helps safeguard their privacy. This complies with current privacy regulations and fosters user trust. It also demonstrates how the system upholds applicants' rights.

- **REQ-5.5:** Breaches must be logged and trigger a notification to the company.

  Rationale: Logging and reporting breaches enables the business to react swiftly to security issues, which is why this criterion is necessary. The company must be informed immediately if sensitive data is at danger so that appropriate action can be taken. This lessens damage, upholds legal requirements, and maintains system security.

3.5.3 Use Cases

UC-5-1: Run Accessibility Audit

An admin performs a system-wide accessibility audit to ensure compliance with WCAG standards and logs any issues into the ticketing/feedback system.

Actors

- **Primary:** Admin or HR
- **Secondary:** Ticketing/Feedback System

Secondary Use Cases

N/A

Preconditions

- The system is running, and user interfaces are active.
- The ticketing/feedback system is connected and available.

Postconditions

- An accessibility report is generated and stored.
- Any detected accessibility issues are automatically logged as tickets in the feedback system.
- A reminder is set for the next audit in 4 weeks.

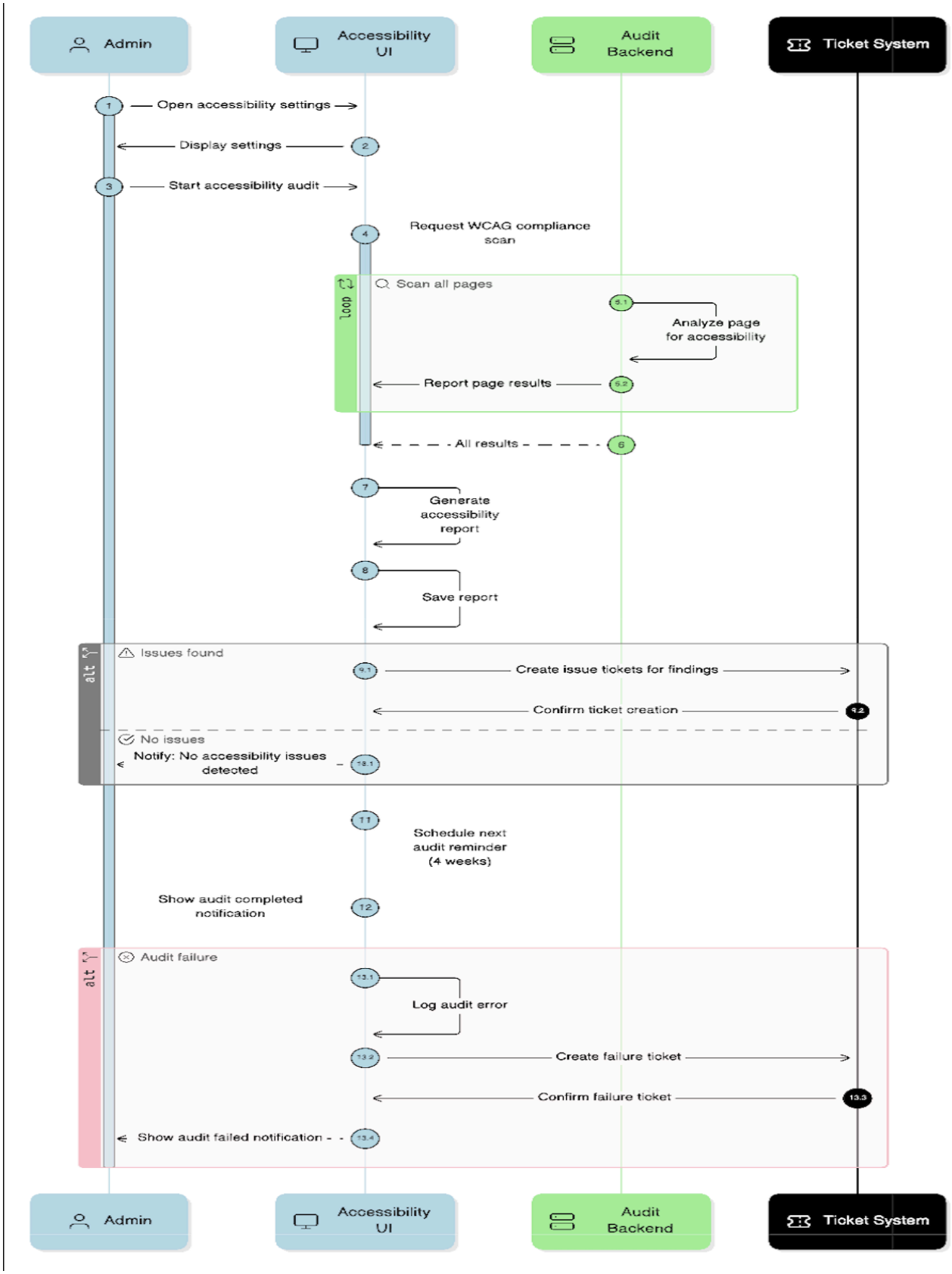
Main Flow

1. The admin navigates to the “Accessibility Settings” tab.
2. The admin clicks the “Run Audit” button.
3. The system scans all pages for accessibility compliance.
4. The system generates an accessibility report.
5. The report is saved for future review.
6. If any accessibility issues are found, the system creates tickets in the ticketing/feedback system with details from the audit.
7. The system sets a reminder for the next audit cycle.

Alternate Flow

At Step 3 of the Main Flow, when the system is checking every page for accessibility compliance, this alternate flow starts. The system promptly reports the error and alerts the administrator if the audit is unsuccessful because of a system error or halted operation. In order to ensure that the failure can be examined and fixed, it then automatically generates a related ticket in the ticketing/feedback system detailing the problem. The audit process is then terminated by the system, and report generation is not carried out.

UC-5-1: Sequence Diagram



UC-5-2: Delete User Data (Right to be Forgotten)

A user requests to delete all their personal and application data from the system. The system also informs the user of the expected date and time when their data will be fully deleted.

Actors

User

Secondary Use Cases

N/A

Preconditions

The user is logged into their account.

Postconditions

- All user data is permanently deleted from the system.
- An audit entry is created confirming the deletion and the exact date/time the deletion occurred.
- The user is shown confirmation including the scheduled deletion date/time.

Main Flow

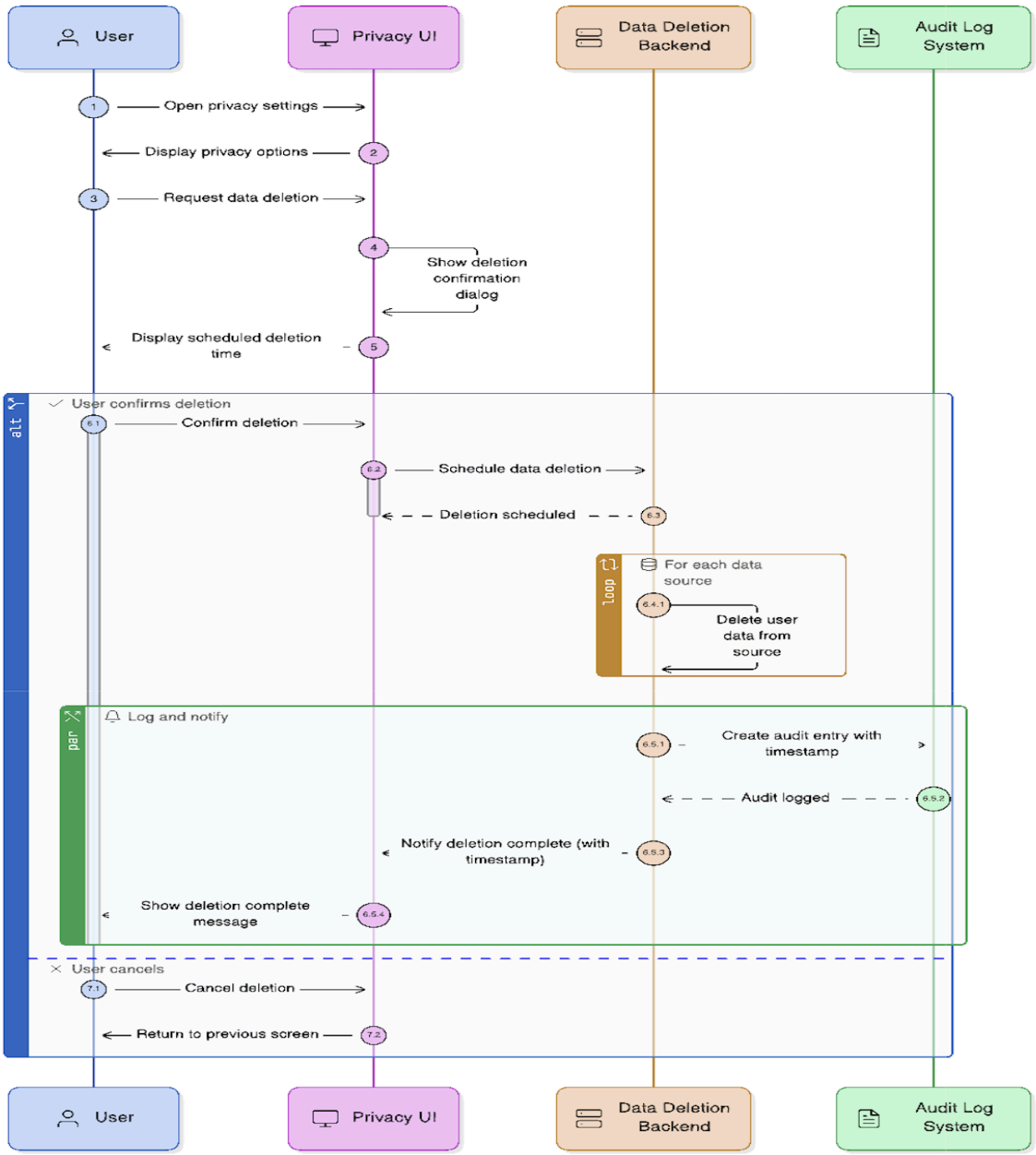
1. The user navigates to the **"Privacy Settings"** section.
2. The user clicks on **"Delete My Data."**
3. The system shows a **confirmation message** explaining that deletion may not happen immediately and displays the **scheduled date and time** for the deletion to occur.
4. The user confirms the deletion request.
5. The system schedules the deletion and begins processing the removal of all stored user data.
6. Once deletion is completed, the system creates an **audit record** showing the exact date and time of deletion.
7. The system sends a **final confirmation message** to the user notifying them that their data was successfully deleted, including the deletion timestamp.

Alternate Flow

If the user cancels the deletion request:

- No data is removed.
- The system returns to the previous screen.

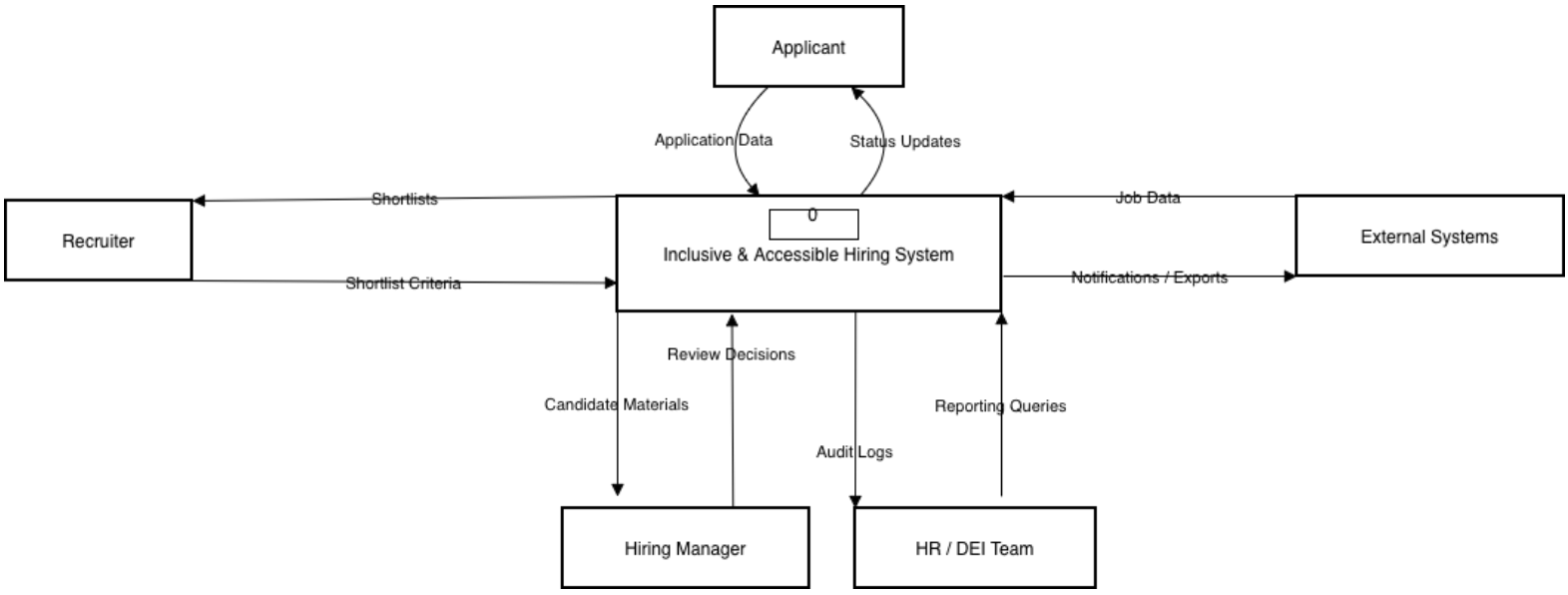
UC-5-2: Sequence Diagram



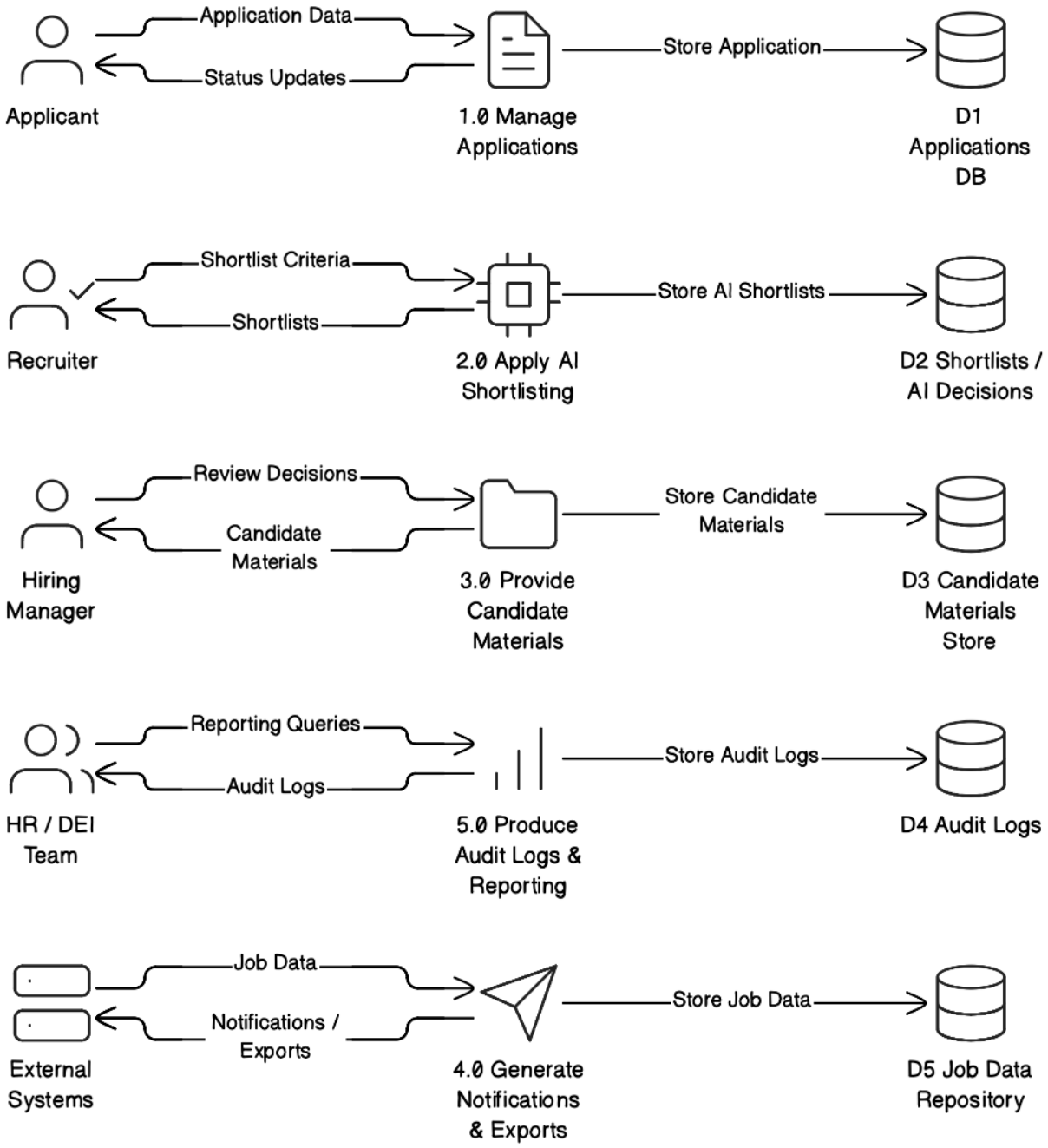
UC-5-2: User Story

As a user, I want to delete all my personal and application data so that I can control my privacy and ensure my information is permanently removed from the system.

3.6 Context Flow Diagram



3.7 System-Wide DFD-0



4.0 External Interface Requirements

4.1 User Interfaces

4.1.1 Description and Priority

The system will provide clear, accessible, and role-specific interfaces tailored to applicants, recruiters, hiring managers, and HR/DEI teams. These interfaces will meet WCAG accessibility standards and ensure compliance with privacy and inclusivity goals. Interfaces will be consistent in layout, navigation, and style to create a seamless experience across devices.

Key design principles:

- Role-based access control for feature-specific UIs.
- Consistent design language and accessibility compliance.
- Clear, descriptive feedback messages and error handling.
- Secure login, session management, and privacy controls such as data encryption, user consent management, and restricted visibility of sensitive information.

Priority: **High**

4.1.2 Functional Requirements

- **REQ-4.1.1:** Applicants shall have screens to upload resumes, preview parsed data, edit skill information, and track application status.

  Rationale: This guarantees that candidates may actively oversee their application process, fix any parsing mistakes in the talents that the AI derived, and monitor the progress of their applications. It contributes to a favorable candidate experience by increasing transparency, decreasing errors from automatic parsing, and improving usability.

- **REQ-4.1.2:** Recruiters shall have dashboards showing candidate shortlists, AI recommendations, one-sentence explanations, and override options.

  Rationale: To make effective selections, recruiters need to see candidate information in a simple and succinct manner. Decision-making is supported while upholding accountability when AI recommendations are accompanied by explanations. Override possibilities handle issues of fairness and trust while guaranteeing recruiters maintain control.

- **REQ-4.1.3:** Hiring managers shall view extended candidate materials such as portfolios, cover letters, and interview notes.

  Rationale: Beyond automatic rankings, hiring managers frequently need more in-depth information about candidates' skills and suitability. Extended material access facilitates comprehensive assessments and well-informed choices, especially for specialist or creative professions.

- **REQ-4.1.4:** HR/DEI teams shall have dashboards for audit logs, diversity metrics, and compliance reports.

  Rationale: Monitoring indicators related to diversity, fairness, and compliance is essential for accountability and governance. Dashboards assist in detecting and addressing possible bias in hiring procedures, promote regulatory compliance, and offer transparency.

- **REQ-4.1.5:** All user interfaces shall be compliant with WCAG and support screen readers, keyboard-only navigation, high contrast mode, and voice input.

  Rationale: All users, including those with disabilities, can utilize the system efficiently thanks to accessibility. In addition to meeting regulatory criteria and reflecting ethical hiring practices, WCAG compliance and assistive features foster inclusion.

- **REQ-4.1.6:** Interfaces shall adapt responsively to desktops, tablets, and mobile devices.

  Rationale: Users use a variety of devices to access the system. Enhancing usability and adoption across platforms, responsive design guarantees a consistent and effective experience.

- **REQ-4.1.7:** All user interfaces shall support English and French at launch, with the capability for additional languages.

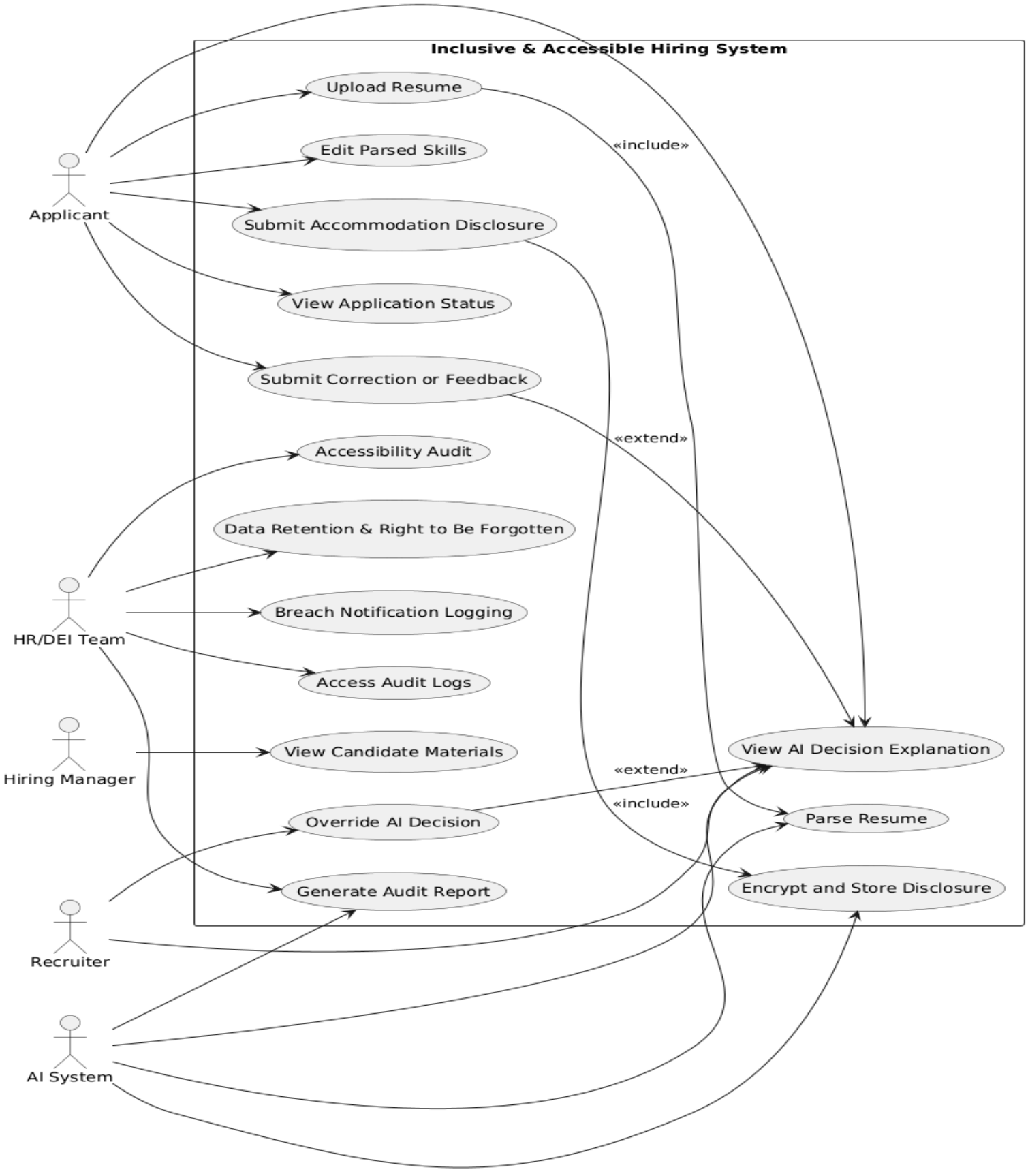
  Rationale: Supporting both official languages guarantees inclusivity for a range of users and complies with local regulations. Future language support is made possible by designing for extensibility without requiring significant redesigns.

4.1.3 Use-Case Diagram

The following Use Case Diagram provides a high-level overview of the primary interactions between the main actors and the Inclusive & Accessible Hiring System. It visually represents all functional requirements identified in **Section 3** of this document, ensuring traceability between the system's features, actors, and their corresponding use cases.

The diagram captures the relationships among **Applicants**, **Recruiters**, **Hiring Managers**, **HR/DEI Teams**, and the internal **AI System**. It illustrates how users engage with system features such as resume parsing, accommodation disclosure, fairness and transparency functions, accessibility auditing, and data-management operations.Each use case in this diagram corresponds directly to at least one functional requirement described in **Sections 3.1 through 3.5**, and all include properly defined pre-conditions, post-conditions, and actor responsibilities as detailed earlier.

Inclusive & Accessible Hiring System - Use Case Diagram



Description of Figure 4-1:
The use case diagram above illustrates the main actors and interactions within the Inclusive & Accessible Hiring System. The diagram includes:

- **Actors:** Applicant, Recruiter, Hiring Manager, HR/DEI Team, and AI System.
- **Use Cases:** Functionalities such as resume upload and parsing, accommodation disclosure, application tracking, AI explainability, auditing, and compliance operations.
- **Relationships:**
    - **Include** (<<include>>): Mandatory sub-flows, e.g., "Upload Resume" includes "Parse Resume".
    - **Extend** (<<extend>>): Optional or conditional flows, e.g., "Submit Correction or Feedback" extends "View AI Decision Explanation".

4.2 Hardware Interfaces

4.2.1 Description and Priority

The system should run on standard user devices, including desktops, laptops, and tablets. It must support common input/output devices like keyboard, mouse, touch screen, and screen readers. The system communicates with the hardware via standard web protocols and must work with devices running modern browsers.

Priority: **Medium**

4.2.2 Functional Requirements.

- **REQ-4.2.1:** System shall detect the device type and adjust the interface layout accordingly.

  Rationale: An ideal user experience on PCs, tablets, and mobile devices is guaranteed by the interface's automatic adaptation. By presenting content in a layout appropriate for the screen size and capabilities of the device, it enhances usability and accessibility.

- **REQ-4.2.2:** System shall accept input from keyboards, touchscreens, voice commands, and alternative input devices.

  Rationale: Diverse user needs, such as those of users with motor disabilities or varied device preferences, are accommodated by supporting alternative input methods. It improves overall system flexibility, inclusivity, and accessibility.

- **REQ-4.2.3:** All interactive controls shall be fully accessible via keyboard and screen readers.

  Rationale: In accordance with WCAG standards and to promote fair access, keyboard and screen reader accessibility guarantees that users with visual or mobility impairments can navigate and engage with the system effectively.

- **REQ-4.2.4:** System shall support low-bandwidth and older hardware without significant performance degradation.

  Rationale: System reach and usefulness are increased by ensuring functionality on slow networks or low-spec devices. It promotes inclusivity and equity by lowering obstacles for users in areas with poor connectivity or outdated equipment.

- **REQ-4.2.5:** System shall integrate with assistive technologies, including JAWS, NVDA, VoiceOver, and Dragon NaturallySpeaking

  Rationale: Integration with widely used assistive technologies enhances accessibility, compliance, and user happiness by enabling people with disabilities to utilize their preferred tools for interaction and navigation.

4.3 Software Interfaces

4.3.1 Description and Priority

The system shall integrate with essential software components, including databases, operating systems, libraries, and internal tools. It must run on modern web browsers and use cloud services for hosting and storage. Data flows include resume files, parsed skills, AI recommendations, and audit logs.

Priority: **High**

4.3.2 Functional Requirements

- **REQ-4.3.1:** System shall use a relational database to store resumes, parsed skills, accommodation disclosures, AI outputs, and audit logs.

  Rationale: Important data, including resumes, parsing skills, and audit logs, can be stored in a relational database in an organized, dependable, and consistent manner. For HR systems that handle sensitive data, it offers robust transaction guarantees, supports sophisticated searches, and guarantees data integrity.

- **REQ-4.3.2:** System shall operate on Linux-based servers and support major modern browsers (Chrome, Edge, Firefox, Safari).

  Rationale: Strong security, stability, scalability, and broad enterprise deployment compatibility are all provided by Linux. Enhancing accessibility and lowering usage barriers, support for all major

browsers guarantees that users may access the system regardless of device or browser preference.

- **REQ-4.3.3:** System shall integrate with NLP/ML models for resume parsing and skill inference, with transparent logs.

  Rationale: Automated skill and pertinent information extraction from resumes is made possible by NLP/ML integration, increasing consistency and efficiency. Ensuring traceability, supporting explainability needs, and aligning with fairness and accountability goals in HR technologies are all made possible by keeping visible records.

- **REQ-4.3.4:** System shall provide secure REST API endpoints for data access and third-party integrations.

  Rationale: With internal modules or external systems like ATS tools, reporting platforms, or external HR services, REST APIs enable safe and consistent data interchange. This preserves robust security measures while enhancing interoperability, scalability, and modular development.

- **REQ-4.3.5:** System shall log all API calls and data access for auditing purposes.

  Rationale: In order to support accountability, compliance (such as GDPR and PIPEDA), and security monitoring, logging guarantees complete traceability of data transactions. It makes it possible to examine questionable activity, verifies system utilization, and offers transparency for fairness and DEI audits.

- **REQ-4.3.6:** System shall support API access for reporting, analytics, and DEI dashboards.

  Rationale: Data retrieval for analytics, reporting, and DEI dashboards is made possible by granting both internal and external tools API-level access. This guarantees that the system may expand into larger HR and DEI activities, enhances organizational insights, and supports data-driven decisions.

4.4 Communications Interfaces

4.4.1 Description and Priority

The system must support web-based communications, including resume uploads, AI processing, notifications, and HR reporting. All communications should be secure, using HTTPS, and compatible with email alerts and browser notifications.

Priority: **High**

4.4.2 Functional Requirements

- **REQ-4.4.1:** System shall enable candidates to securely upload resumes via HTTPS.

  Rationale: Resume files and personal information are encrypted during transmission when HTTPS is used. This promotes adherence to privacy laws, shields private application data from interception, and builds confidence in the platform's security.

- **REQ-4.4.2:** System shall send email and in-app notifications to candidates and recruiters for application status updates.

  Rationale: Important events like status updates, interview schedules, or new submissions are sent to all parties through notifications. This guarantees that users remain involved throughout the recruiting process, encourages transparency, and enhances communication.

- **REQ-4.4.3:** System shall communicate with databases, AI modules, and external APIs via secure protocols.

  Rationale: Data transferred between system components and outside services is protected by secure communication protocols. This guarantees adherence to security best practices throughout the platform's architecture, stops unwanted access, and maintains data integrity.

- **REQ-4.4.4:** System shall support export of reports in formats such as PDF, CSV, and Excel.

  Rationale: HR, DEI teams, and recruiters can use their preferred tools to evaluate data thanks to the support for numerous export formats. It enables audit, compliance, and data-driven decision-making while also improving interoperability and streamlining external reporting operations.

- **REQ-4.4.5:** System shall ensure real-time updates are reflected across all user interfaces.

  Rationale: Users will always view the most recent information, such as newly generated recommendations or updated application statuses, thanks to real-time synchronization. This avoids misunderstandings, lowers mistakes, and enhances the platform's general dependability.

- **REQ-4.4.6:** System shall provide secure webhook or API-based notifications for integrated systems.

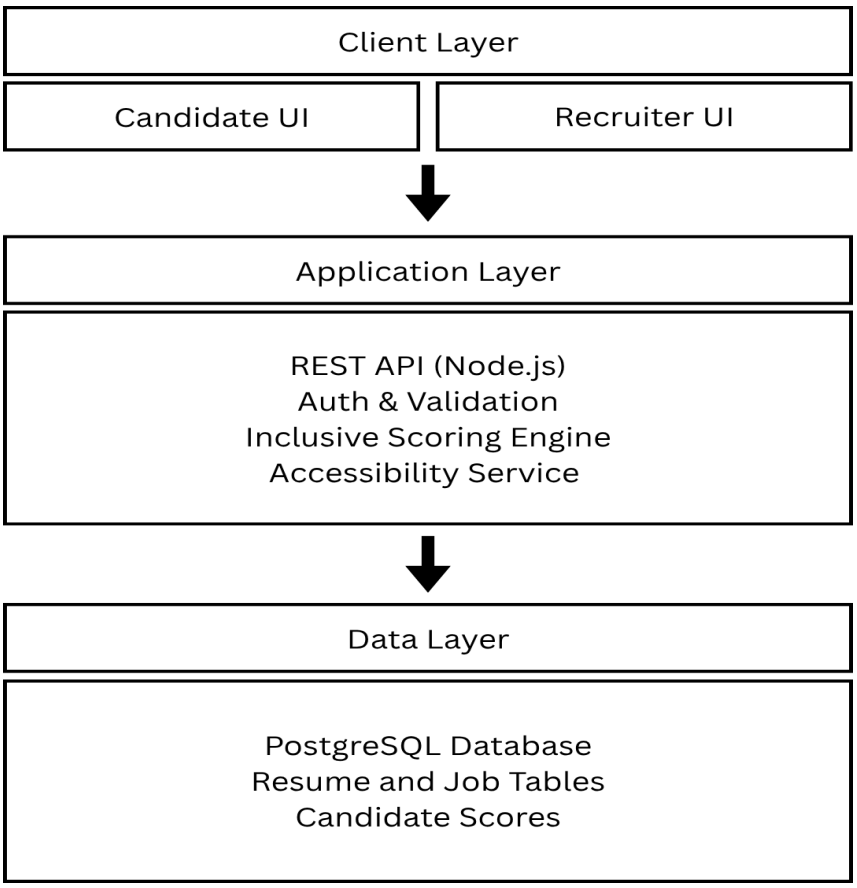
  Rationale: Easy contact with third-party tools like ATS systems, reporting platforms, or HR workflows is made possible using webhooks and API-based notifications. This facilitates scalable ecosystem connectivity, enhances system integration, and automates external operations.

- **REQ-4.4.7:** System shall log all communication events for traceability and auditing.

Rationale: Every status update, API call, and notification may be tracked thanks to the logging of communication events. Maintaining a transparent operational history within the system, security audits, compliance verification, and dispute resolution all depend on this.

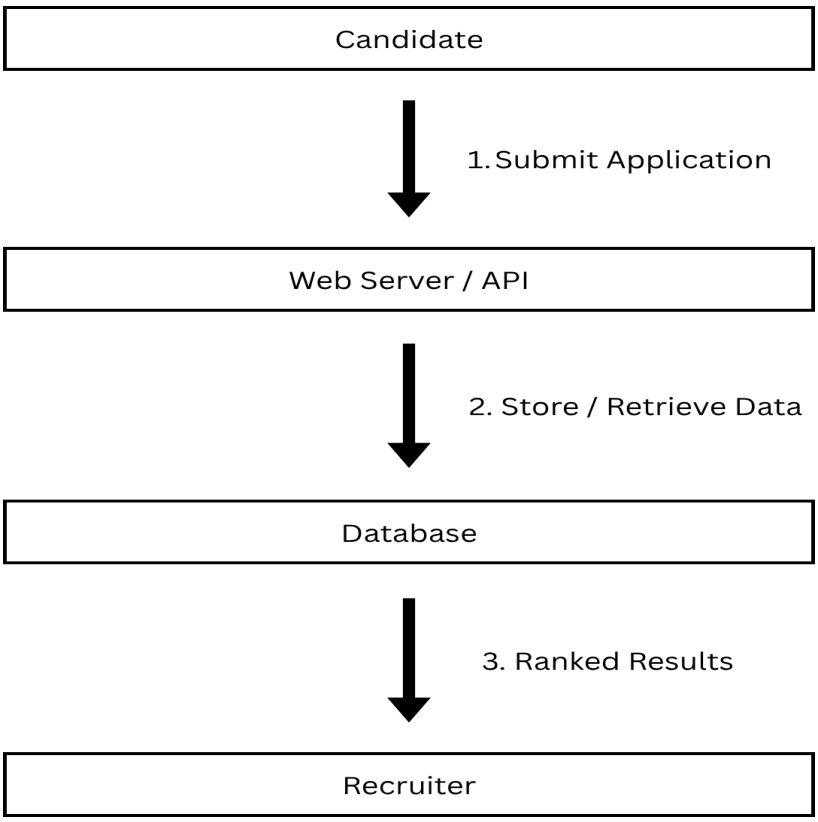
4.5 Interface Diagrams

Figure 4-1: System Context Diagram



Illustrates the interaction between applicants, recruiters, hiring managers, HR/DEI teams, ATS, AI/NLP modules, and databases.

Figure 4-2: Data Flow Diagram



Shows the flow of data between interfaces, including resume uploads, skill parsing, accommodation disclosure, candidate evaluation, and audit logging.

5.0 Other Non-Functional Requirements

5.1 Performance Requirements

5.1.1 Description and Priority

All system features must meet specific performance requirements to ensure work is completed quickly and efficiently. The system must be available at all times, have a quick response time, scale well to support a large number of users, and be reliable to use.

Priority: **Medium**

5.1.2 Non-Functional Requirements

- **REQ-5.1.1:** The application should be available 99.9% of the time over 12 months with an acceptable downtime threshold of 8 hours and 46 minutes per year. Any reports of the system failing should be investigated and fixed immediately.

  Rationale: A high availability is essential for ensuring users and recruiters can access the application at any time, this is especially important due to how volatile the job market is.

- **REQ-5.1.2:** The system should load the main dashboard and any additional windows in under a second.

  Rationale: Fast load times improve the user experience and help avoid frustration.

- **REQ-5.1.3:** The system must be able to support 50,000 concurrent visits at a single time without any noticeable delay longer than 100ms.

  Rationale: Supporting large-scale concurrent usage ensures the system remains stable during peak traffic.

- **REQ-5.1.4:** All application features should be highly reliable, maintaining full functionality throughout 99.9% of the year.

  Rationale: High reliability ensures that all features work as intended throughout the year, this protects user trust and ensures the platform supports quick, efficient work.

5.2 Safety Requirements

5.2.1 Description and Priority

To ensure the safety of our users, multiple safeguards and policies should be established to prevent an incident from occurring, and the company cannot be held liable if one does. The system should validate correct data, purge data collected after a certain time, and ensure that there are proper measures to ensure that info is saved only when prompted.

Priority: **High**

5.2.2 Non-Functional Requirements

- **REQ-5.2.1:** The interface must automatically validate that all required information fields are accurate and in the proper form.

  Rationale: Automatically validating user inputs prevents errors, incorrect data processing, and reduces the likelihood of system misuse

- **REQ-5.2.2:** The system must delete any stored data after a maximum of five years.

  Rationale: Deleting data after five years helps comply with data retention policies and privacy regulations. It also minimizes data breaches and prevents unnecessary data being left in the system.

- **REQ-5.2.3:** Users must have the option to request the removal of their info from the system 100% of the time during the year. After a request, the info must be wiped from the system within 30 days of the request.
  Rationale: Giving users full control over the removal of their information supports privacy rights, and ensuring deletion within 30 days lets employees get to it within a reasonable time without completely forgetting about it.

- **REQ-5.2.4:** The user must be informed of what information will not be saved by the system, when applicable.

  Rationale: informing users about which data the system does not save prevents misunderstandings. This also helps protect the organization from liability and establishes clearer user expectations in terms of data handling.

5.3 Security Requirements

5.3.1 Description and Priority

The system must create security measures to protect the confidentiality, integrity, and availability of all company data, recruiter information, and personal identifying information. This includes protecting data, ensuring all requirements comply with GDPR (General Data Protection Regulation), and logging all security-relevant events.

Priority: **High**

5.3.2 Non-Functional Requirements

- **REQ-5.3.1:** All user accounts should require multi-factor authentication.

  Rationale: Multi-factor authentication provides an additional layer of protection that reduces the chance of unauthorized account access.

- **REQ-5.3.2:** Passwords must have a minimum complexity policy of 10 characters to ensure that brute-forcing a correct password would be impossible.

  Rationale: A strong password makes brute forcing a password significantly harder to reduce the chance of unauthorized account access.

- **REQ-5.3.3:** After 5 wrong password attempts, the user must not be able to try again for 10 minutes.

  Rationale: Rate-limiting login attempts prevents brute forcing a password and reduces the likelihood of account compromise. A 10-minute lockout provides balance between security and user convenience.

- **REQ-5.3.4:** Access to all data and functionality shall be governed by Role-Based Access Control (RBAC).

  Rationale: Role-Based Access Control ensures users only access information and features relevant to their role.

- **REQ-5.3.5:** The system must include an immutable audit log of all security-related events such as password attempts, data access and user permissions.

  Rationale: An audit log allows tracking of all critical security related events, which can lead to rapid response to any alarming actions and irrefutable evidence of it occurring.

5.4 Software Quality Attributes

5.4.1 Description and Priority

The system must prioritize a high degree of quality to ensure users and developers can efficiently manage their workflow without system interruptions. To achieve this, the app should be easily maintainable, have low-cost adaptation to changing laws, and be able to have the core components reusable for future endeavours.

Priority: **Low**

5.4.2 Non-Functional Requirements

- **REQ-5.4.1:** A new user should be able to use and understand how to upload their resumes, apply to jobs, and manage their profile with less than 30 minutes of training. A new recruiter should be able to use the filtering features, understand the data presented, and be able to accept resumes within the same time.

  Rationale: Ensuring the interface is intuitive reduces onboarding time, increases user satisfaction, and lowers training costs.

- **REQ-5.4.2:** The code structure must allow for patches to be done while the system is live.

  Rationale: Allowing live patches minimizes downtime and ensures critical updates can be applied without interrupting users.

- **REQ-5.4.3:** Any security patches must be able to be implemented in less than four hours after identification.

  Rationale: Fast deployment of security patches reduces exposure time to vulnerabilities, minimizing the risk of breaches.

- **REQ-5.4.4:** The core structure of the software should be localized into components that can be easily reused in future projects.
  Rationale: Reusable components reduce development time for future projects and make maintenance easier.

- **REQ-5.4.5:** All documentation should be up to date and thorough.

  Rationale: Thorough and up-to-date documentation ensures developers and administrators can maintain, update, and troubleshoot the system effectively.

6.0 Other Requirements

6.1 Database Requirements

- **REQ-6.1:** The system must store candidate resumes, accommodations data, AI outputs (such as skill inferences), and audit logs in a relational database.

  Rationale: This requirement ensures that all critical hiring data is stored in a structured, queryable format, making it easier to support reporting, auditing, and future feature changes.

- **REQ-6.2:** All stored data must be encrypted at rest and in transit, ensuring compliance with relevant data privacy laws.

  Rationale: This requirement protects sensitive personal information from unauthorized access or interception, reducing security risks and supporting legal compliance obligations.

- **REQ-6.3:** Data retention must be limited to 5 years, after which data must be securely deleted, in compliance with the organization's retention policies.

  Rationale: This requirement prevents indefinite storage of personal data, lowering legal and security exposure while aligning the system with organizational and regulatory retention rules.

- **REQ-6.4:** The system must allow easy retrieval of candidate data by authorized users (e.g., recruiters, HR personnel) for review and audit purposes.

  Rationale: This requirement ensures that staff can efficiently access relevant records to support hiring decisions, respond to audits, and investigate issues without complex manual work.

6.2 Internationalization Requirements

- **REQ-6.5:** The system must support English and French at launch, with the potential to add additional languages in the future.

  Rationale: This requirement ensures that the platform is usable by speakers of both official languages and that the architecture can scale to additional languages without major redesign.

- **REQ-6.6:** The system must comply with local accessibility regulations.

Rationale: This requirement guarantees that the system meets jurisdiction-specific legal obligations, reducing compliance risk and supporting equitable access for all users.

6.3 Legal Requirements

- **REQ-6.7:** The system must ensure data privacy and security in compliance with applicable laws.

  Rationale: This requirement ties the technical design directly to legal standards, helping the organization avoid fines, reputational damage, and legal action related to mishandling personal data.

- **REQ-6.8:** Sensitive personal information, such as accommodations, must be stored separately from resumes, encrypted, and excluded from influencing AI rankings.

  Rationale: This requirement prevents discrimination by ensuring that accommodation details cannot bias AI or human decisions, while still keeping the information available for legitimate accessibility and HR purposes.

6.4 Reuse Objectives

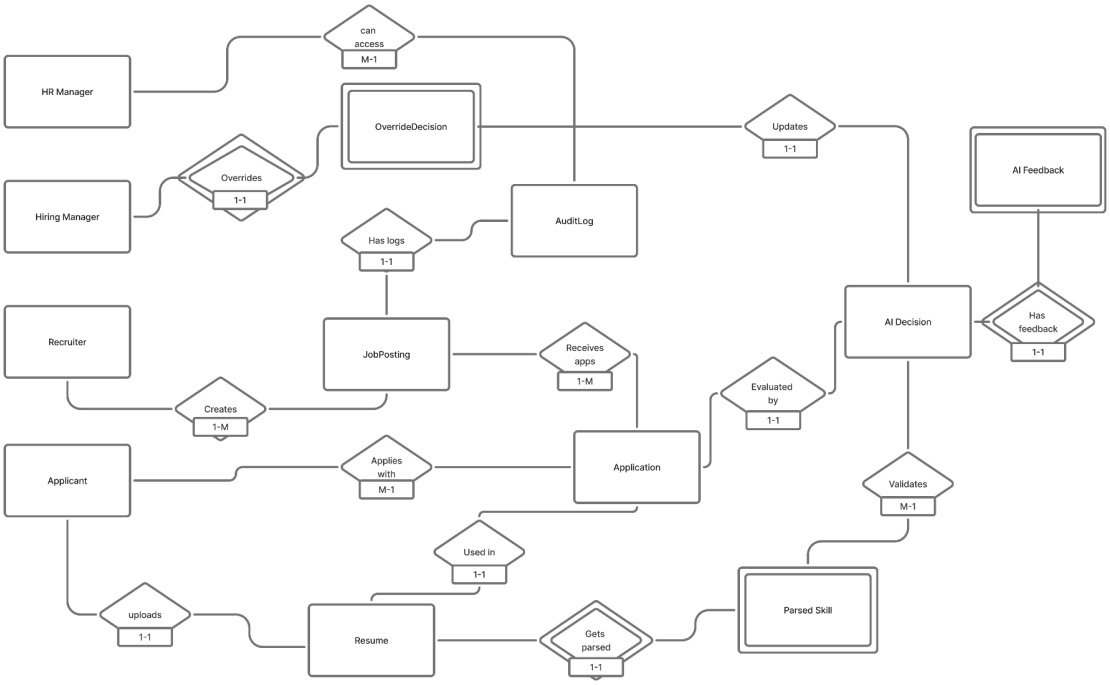
- **REQ-6.9:** Reuse of existing AI models for resume parsing and skills inference is encouraged to minimize development time and enhance system reliability.

  Rationale: This requirement leverages proven AI components, reducing implementation risk, shortening time-to-delivery, and allowing the team to focus effort on fairness, accessibility, and integration work.

7.0 Entity Relationship Diagram

This section describes the relationships between existing features, users, and objects included in the project.

7.1 Entity Relationship Diagram



7.2 Data Dictionary

Applicant:

Attribute	Type	Description
applicant_id (PK)	UUID	Unique identifier for an applicant.
name	VARCHAR	Applicant’s full name.
email	VARCHAR	Applicant’s contact email.
phone	VARCHAR	Applicant’s phone number (optional).
account_created	TIMESTAMP	Date the applicant’s account was created.

Recruiter:

Attribute	Type	Description
recruiter_id (PK)	UUID	Unique identifier for a recruiter.
name	VARCHAR	Recruiter’s full name.

email	VARCHAR	Recruiter's email.
role	VARCHAR	Role assigned (Recruiter).

Hiring Manager:

Attribute	**Type**	**Description**
hiring_manager_id (PK)	UUID	Unique ID for hiring manager.
name	VARCHAR	Manager's name.
email	VARCHAR	Manager email.
department	VARCHAR	Department they manage.

HRManager:

Attribute	**Type**	**Description**
hr_manager_id (PK)	UUID	Unique ID for HR/DEI staff member.
name	VARCHAR	HR staff name.
email	VARCHAR	Work email.
role	VARCHAR	HR/DEI role.

Resume:

Attribute	**Type**	**Description**
resume_id (PK)	UUID	Unique identifier for a resume.
applicant_id (FK)	UUID	Applicant who uploaded the resume.
upload_date	TIMESTAMP	Date the resume was uploaded.
file_url	VARCHAR	Path to stored resume file.

ParsedSkill:

Attribute	**Type**	**Description**
parsed_skill_id (PK)	UUID	Unique ID for parsed skill entry.
resume_id (FK)	UUID	Resume from which the skill was parsed.
skill_name	VARCHAR	Extracted skill.
confidence_score	DECIMAL	AI confidence score for extraction.

JobPosting:

Attribute	Type	Description
job_posting_id (PK)	UUID	Unique posting ID.
recruiter_id (FK)	UUID	Recruiter who created the job posting.
title	VARCHAR	Job title.
description	TEXT	Description of the role.
created_date	TIMESTAMP	When the posting was created.

Application:

Attribute	Type	Description
application_id (PK)	UUID	Unique application ID.
applicant_id (FK)	UUID	Applicant submitting application.
job_posting_id (FK)	UUID	Job posting applied to.
resume_id (FK)	UUID	Resume used.
submission_date	TIMESTAMP	When the applicant applied.

AIDecision:

Attribute	Type	Description
ai_decision_id (PK)	UUID	Unique identifier for AI evaluation result.
application_id (FK)	UUID	Application being evaluated.
decision_outcome	VARCHAR	Example: shortlisted / rejected.
explanation	TEXT	One-sentence explanation (XAI).
decision_date	TIMESTAMP	When AI generated the decision.

AIFeedback:

Attribute	Type	Description
ai_feedback_id (PK)	UUID	Unique identifier for user feedback.
ai_decision_id (FK)	UUID	AI decision receiving feedback.

feedback_text	TEXT	Correction or feedback submitted.
submitted_by_applicant	BOOLEAN	Whether feedback was given by the applicant.

OverrideDecision:

Attribute	**Type**	**Description**
override_id (PK)	UUID	Unique ID for override record.
hiring_manager_id (FK)	UUID	Manager performing the override.
ai_decision_id (FK)	UUID	Decision being overridden.
justification	TEXT	Reason for the override.
override_date	TIMESTAMP	When override happened.

AuditLog:

Attribute	**Type**	**Description**
audit_id (PK)	UUID	Unique audit log entry.
related_entity	VARCHAR	“AI Decision”, “Override”, “Disclosure”, etc.
related_id	UUID	ID of the target record.
action	VARCHAR	What happened (Viewed, Updated, Overridden, etc).
timestamp	TIMESTAMP	When the event was logged.
hr_manager_id (FK, nullable)	UUID	HR user who accessed logs (optional).

FOREIGN KEYS DICTIONARY

Foreign Key	**Source Table.Column**	**References Table.Column**
FK_Resume_Applicant	Resume.applicant_id	Applicant.applicant_id
FK_ParsedSkill_Resume	ParsedSkill.resume_id	Resume.resume_id
FK_JobPosting_Recruiter	JobPosting.recruiter_id	Recruiter.recruiter_id
FK_Application_Applicant	Application.applicant_id	Applicant.applicant_id
FK_Application_JobPosting	Application.job_posting_id	JobPosting.job_posting_id
FK_Application_Resume	Application.resume_id	Resume.resume_id
FK_AIDecision_Application	AIDecision.application_id	Application.application_id
FK_AIFeedback_AIDecision	AIFeedback.ai_decision_id	AIDecision.ai_decision_id

FK_OverrideDecision_Manager	OverrideDecision.hiring_manager_id	HiringManager.hiring_manager_id
FK_OverrideDecision_AIDecision	OverrideDecision.ai_decision_id	AIDecision.ai_decision_id
FK_AuditLog_HRManager	AuditLog.hr_manager_id	HRManager.hr_manager_id

UNIQUE KEYS DICTIONARY

Unique Key	**Table**	**Column(s)**	**Reason**
UQ_Applicant_Email	Applicant	email	Prevent duplicate applicant accounts.
UQ_Recruiter_Email	Recruiter	email	Ensure one unique recruiter login per email.
UQ_HiringManager_Email	HiringManager	email	Avoid duplicates.
UQ_HRManager_Email	HRManager	email	Avoid duplicates.
UQ_Resume_FileURL	Resume	file_url	Prevent storing the same uploaded file twice.
UQ_ParsedSkill_SkillPerResume	ParsedSkill	resume_id, skill_name	Prevent duplicate skill entries for the same resume.
UQ_Application_OnePerPosting	Application	applicant_id, job_posting_id	Prevent applying multiple times to the same posting.
UQ_AIDecision_PerApplication	AIDecision	application_id	Only one AI decision per application.
UQ_AIFeedback_OnePerDecision	AIFeedback	ai_decision_id	One feedback record per AI decision
UQ_OverrideDecision_OnePerDecision	OverrideDecision	ai_decision_id	One final override state,
UQ_AuditLog_PK	AuditLog	audit_id	Ensures audit logs are unique.

8.0 Appendix: Issues List

8.1 TBDs

- **TBD-1:** Voluntary disclosure design (timing disclaimer; consent for storage vs. audit/reporting).

8.2 Information Needed

- **IN-1:** Accepted accommodation categories and secure storage/display rules (job-critical vs. non-critical).

8.3 Conflicts Awaiting Resolution

- **CR-1:** Stage-based anonymization policy (Initial: fully anonymous; Final: reveal only job-relevant constraints).